

From Risk Registers to Uncertainty Observatories: Managing Shadow Generative AI Innovation Under High Risk and High Uncertainty

Abstract

Generative AI has made digital innovation easier to initiate, harder to observe, and harder to govern with conventional risk registers. This paper develops shadow generative AI innovation as work-use of GenAI systems, agents, or model-enabled workflows that can create organizational value while remaining only partly authorized, visible, instrumented, or accountable. Building on shadow IT, shadow AI, shadow user innovation, digital innovation, and dynamic capabilities research, we argue that the managerial problem is not simply high risk, where hazards can be categorized in advance, but high uncertainty, where actors, tasks, tools, and consequences change faster than governance categories stabilize. A mechanism-oriented public evidence map combines AI governance and digital innovation metadata, the U.S. Census BTOS AI supplement, Stack Overflow Developer Survey 2025, Anthropic Economic Index task data, and public AI Status Dashboard provider records. The evidence is interpreted as mechanism-oriented proxy evidence rather than direct measurement of hidden conduct. It maps conditions consistent with an adoption-governance gap: 17.9% of U.S. employer businesses report current AI use in any function, 78.5% of Stack Overflow respondents use AI tools, but only 32.8% report positive trust in AI-output accuracy and only 27.3% of AI-agent users report observability or security tooling. We introduce uncertainty observatories as a governance capability that complements risk registers through weak-signal sensing, safe disclosure, boundary mapping, containment, reversibility analysis, and institutional learning. The contribution is a proposition-driven framework for managing GenAI innovation under VUCA conditions when consequential uses may emerge outside formal innovation portfolios.

Keywords: Generative AI, Digital innovation, VUCA, Uncertainty, Risk governance, Shadow IT, Dynamic capabilities

1. Introduction

Generative artificial intelligence (GenAI) has changed the practical conditions under which digital innovation begins inside organizations (Krishna et al., 2024). Earlier waves of enterprise digital transformation typically required procurement, integration, data engineering, business-case approval, or explicit platform access before a new technology could become materially embedded in work (Kanitz et al., 2023). GenAI lowers many of these thresholds. A project manager can use a general-purpose model to draft a client memo, a software developer can ask an agent to refactor code and open a pull request, a marketing analyst can generate campaign variants, and a support team can summarize sensitive customer exchanges before the organization has decided whether these activities are approved, logged, auditable, or aligned with policy. The resulting innovation can be valuable. It can accelerate learning, reduce bottlenecks, increase local experimentation, and democratize capabilities long concentrated in specialist teams. Yet the same features that make GenAI generative also make it difficult to govern: prompts are ephemeral, outputs are probabilistic, models change across releases, agentic tools call other tools, and employees can combine public services, enterprise copilots, browser extensions, local models, and workflow automations in ways that leave weak traces in formal systems (Oswick, 2024).

This paper addresses that governance problem by developing the concept of *shadow generative AI innovation*. The term is deliberately narrower than all AI use and broader than malicious violation. It refers to work-use of GenAI systems, agents, or model-enabled workflows that creates or seeks to create organizational value while remaining only partly authorized, visible, instrumented, or accountable. The conceptual problem is familiar in one sense: organizations have long dealt with shadow IT, end-user computing, and locally improvised systems (Behrens, 2009; Furstenau et al., 2017; Silic and Back, 2014). Recent work has now brought the shadow-AI problem closer to the center of technology-and-society scholarship. Chin et al. (2025) analyze shadow AI usage in metaverse-based business models and show why it has paradoxical consequences for organizational knowledge leakage and sustainability. Waters-Lynch et al. (2025) theorize shadow user innovation as covert GenAI-enabled value creation that can be disclosed and converted into dynamic-capability renewal. These studies are important predecessors rather

than targets to be displaced. The present paper builds from them but shifts the object of analysis from shadow AI usage or covert user innovation to partially visible task chains under uncertainty. The problem is not only whether employees use unapproved AI or whether they disclose hidden experiments; it is how organizations learn when approved tools, unapproved services, task reliance, data boundaries, output verification, and agentic action combine before stable governance categories exist.

Shadow GenAI innovation is a socio-technical phenomenon in which worker agency, organizational control, professional norms, vendor infrastructures, public policy, and societal expectations about responsible AI meet in everyday work. It is also a timely problem for digital innovation under volatile, uncertain, complex, and ambiguous (VUCA) conditions (Bennett and Lemoine, 2014; Millar et al., 2018). Digital innovation research has emphasized generativity, recombination, layered modular architectures, and the distributed agency of users and developers (Yoo et al., 2010; Nambisan et al., 2017). GenAI intensifies these properties because innovation can begin as a conversation with a model, a chain of prompts, or an agentic workflow. In that setting, the central managerial question is not whether organizations should allow or ban GenAI. The more difficult question is how they can preserve local value creation while identifying when experimentation crosses thresholds of data sensitivity, task criticality, output reliance, irreversibility, or agentic autonomy. Figure 1 presents the problem diagnosis framing shadow GenAI uncertainty as the combined result of local work pressure, tool concealability, verification burden, ambiguous data boundaries, agentic boundary crossing, and governance category lag.

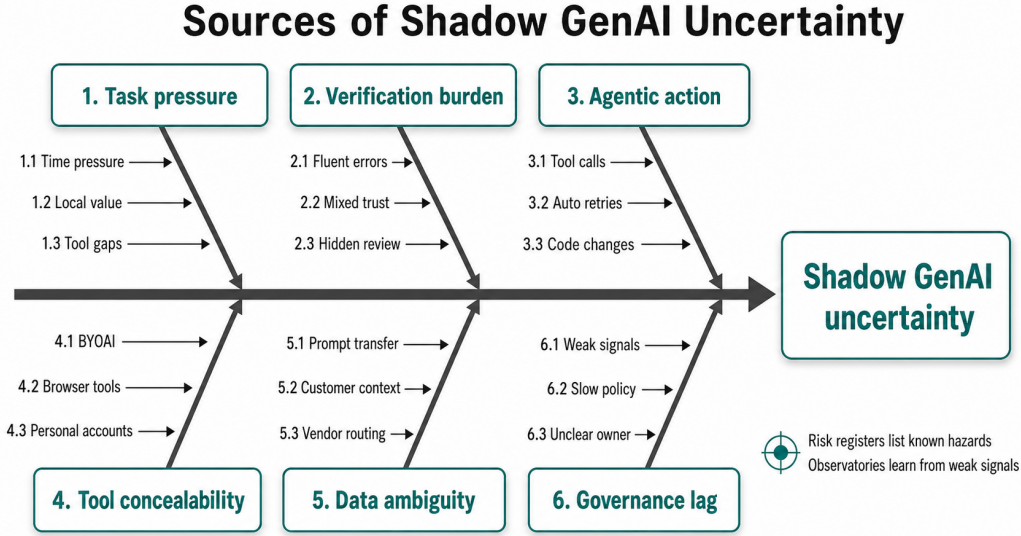


Figure 1: Fishbone diagnosis of shadow GenAI uncertainty. The diagram treats hidden or partly visible GenAI use as a combined product of task pressure, tool concealability, verification burden, data-boundary ambiguity, agentic boundary crossing, and governance category lag.

Conventional AI governance often starts from risk management. Risk registers, model inventories, approval gates, vendor reviews, acceptable-use policies, and compliance checklists remain necessary. They help organizations list known hazards, assign owners, specify controls, and show due diligence. However, they are insufficient when the technology is used unofficially, when approved use changes faster than the register, or when the relevant issue is not a known hazard but a weak signal. A risk register can record the risk that confidential data are entered into an external model. It is less able to capture a new pattern in which employees use an approved coding assistant to generate tests, paste failing snippets into an unapproved chat tool, ask a browser agent to search documentation, and then use an autonomous workflow to alter deployment scripts. Each step can look locally reasonable. The aggregate workflow can be poorly observed, difficult to reverse, and hard to assign to a clear accountable owner.

The theoretical move in this paper is therefore to distinguish *risk-register governance* from *uncertainty-observatory governance*. Risk registers presume that categories can be stabilized before action. Uncertainty observatories presume that categories often become visible only through distributed sensing,

safe disclosure, anomaly capture, and institutional learning. The observatory metaphor is not meant to imply employee surveillance. It describes a governance capability for seeing weak signals without extinguishing useful experimentation: task-level disclosure pathways, model and agent boundary maps, incident and near-miss capture, reversibility assessment, exception review, and learning loops that update policy as uses evolve. The goal is to govern shadow GenAI innovation as a dynamic capability under uncertainty (Teece, 2007; Teece et al., 2016), not as a static list of violations.

Empirically, direct measurement of shadow GenAI is difficult because hidden or partly authorized use is, by definition, not fully visible in public datasets. We therefore use a mechanism-oriented public evidence map that treats public evidence as mechanism-oriented proxy evidence rather than as a census of hidden behavior. The paper combines five evidence streams. First, an author-conducted structured literature synthesis identifies how current AI governance, GenAI, and organizational-work literatures connect responsible AI to digital innovation and where shadow use remains dispersed across concepts. Second, the 2026 U.S. Census Business Trends and Outlook Survey (BTOS) AI supplement provides representative firm-level evidence of AI diffusion by sector, function, and firm size (U.S. Census Bureau, 2026b,a). Third, the Stack Overflow Developer Survey 2025 provides individual-level evidence on AI use, trust, agents, friction, and governance-related tooling among developers (Stack Overflow, 2025). Fourth, the Anthropic Economic Index provides occupation and task exposure signals that help identify where GenAI pressure may concentrate (Anthropic, 2026). Fifth, public AI Status Dashboard provider-status incident data supply operational weak signals about boundary fragility, partial degradation, and technical opacity.

The resulting evidence is consistent with an adoption-governance gap, but it should be read carefully. In the BTOS supplement, 17.9% of U.S. employer businesses report current AI use in any business function, with 51 - Information at 37.6% and the largest employment-size class at 30.6%. In Stack Overflow 2025, 78.5% of valid respondents report current AI-tool use and 64.8% report weekly or daily use. Yet only 32.8% report positive trust in AI-output accuracy, and only 27.3% of AI-agent users report observability or security tooling. The models deepen this pattern. Daily AI use is strongly associated with positive AI-output trust (OR=2.84) and with agent use (OR=3.45), while orchestration tools are strongly associated with observability/security stack use among agent users (OR=4.67). These public data do not measure unauthorized behavior. They identify conditions under which shadow GenAI

becomes plausible and governance-relevant: high adoption, mixed trust, task pressure, agentic tooling, and uneven instrumentation.

To keep this motivation empirical but bounded, Table 1 consolidates selected public and scholarly signals that are consistent with an adoption–governance gap without treating any one source as a direct measure of hidden conduct. Practitioner and industry statistics that further contextualize the problem are reported separately in Section Appendix D. These sources are used only to motivate the practical salience of the problem; they are not part of the three-study empirical design.

Table 1: Selected public and scholarly signals motivating the adoption–governance gap

Signal domain	Public or scholarly indicator	Base	Key statistic	Boundary for this paper
Shadow AI research	Shadow AI usage and organizational knowledge leakage in metaverse-based business models	Survey-based Technology in Society study	Inverted U-shaped relation with knowledge leakage; negative link with business-model sustainability	Establishes shadow AI as an adjacent empirical phenomenon, not this paper’s task-chain construct
Firm AI diffusion	Current AI use by U.S. employer businesses	Census BTOS AI supplement	17.9% current AI use; 37.6% in Information; 30.6% in largest size class	Measures adoption pressure, not hidden employee conduct
Developer work practice	AI-tool use, trust, and agent observability	Stack Overflow 2025 valid responses	78.5% current AI use; 64.8% weekly/daily use; 32.8% positive trust; 27.3% agent observability/security tools	Measures a technically salient worker population, not all employees
Task exposure	Occupation and task concentration of AI exposure	Anthropic Economic Index public data	Mean occupation exposure 7.7%; highest local-file occupation exposure 74.5%	Identifies task pressure, not authorization status or intent
Operational weak signals	Provider-status incidents and degradation patterns	AI Status Dashboard feed, n=25	20 partial-degradation signals; 16 API availability/error incidents	Signals infrastructural fragility, not internal workaround prevalence

The table is a bounded motivation device. It uses public and scholarly signals to show why the adoption–governance gap is practically and theoretically salient, but none of these rows is treated as direct measurement of hidden conduct. Broader practitioner statistics are reported in Section Appendix D.

The paper asks three research questions:

1. How does GenAI change shadow digital innovation from a problem of unauthorized systems into a problem of partially visible, task-level, and agentic work practices?
2. What public empirical signals are consistent with an adoption-governance

gap under high risk and high uncertainty?

3. What governance capability can complement risk registers when GenAI use is valuable, distributed, and only partly observable?

Table 2 links these questions to the three-study design and the article’s contribution logic.

Table 2: Research questions, studies, evidence, and contributions

RQ	Question focus	Primary evidence	Inference boundary	Contribution
RQ1	How GenAI changes shadow digital innovation	Theory plus Study 1 structured literature synthesis and closest-predecessor comparison	Conceptual development, not prevalence measurement	Defines shadow GenAI as task-chain uncertainty rather than merely unauthorized tool use
RQ2	What public signals are consistent with an adoption-governance gap	BTOS, Stack Overflow, Anthropic, and provider-status incident data	Proxy and associational evidence, not direct hidden-behavior logs	Shows conditions under which shadow pressure becomes plausible
RQ3	What governance capability complements risk registers	Cross-study synthesis and framework development	Normative and design-oriented propositions for future testing	Specifies uncertainty observatories as a learning-oriented governance architecture

The roadmap keeps the empirical contribution disciplined: public data ground mechanisms and boundary conditions; the framework translates those mechanisms into governance design.

The paper makes three contributions. First, it extends recent shadow AI and shadow user innovation research by theorizing shadow GenAI as task-chain uncertainty rather than only as unapproved tool use or covert employee innovation. Second, it provides a transparent mechanism-oriented evidence map for the adoption-governance gap using public datasets and reproducible analysis scripts, while stating what those sources cannot support. Third, it proposes uncertainty observatories as an operating governance architecture for weak-signal sensing, safe task disclosure, boundary mapping, containment, reversibility, worker legitimacy, and institutional learning. Across these contributions, GenAI is treated both as an object of innovation and as an instrument through which other innovation is produced. It can create value, leave value missing when useful experimentation is suppressed, and generate value slippage when unobserved use becomes dependency, exposure, or mistrust.

Figure 2 summarizes the architecture of GenAI-enabled innovation: GenAI affordances lower the cost of local experimentation, task concealability and

verification burden make practice only partly visible, and boundary crossing can turn useful experimentation into organizational uncertainty. The figure therefore connects the paper’s problem diagnosis to its governance response: risk registers cover known hazards, whereas uncertainty observatories support weak-signal learning when categories are still forming.

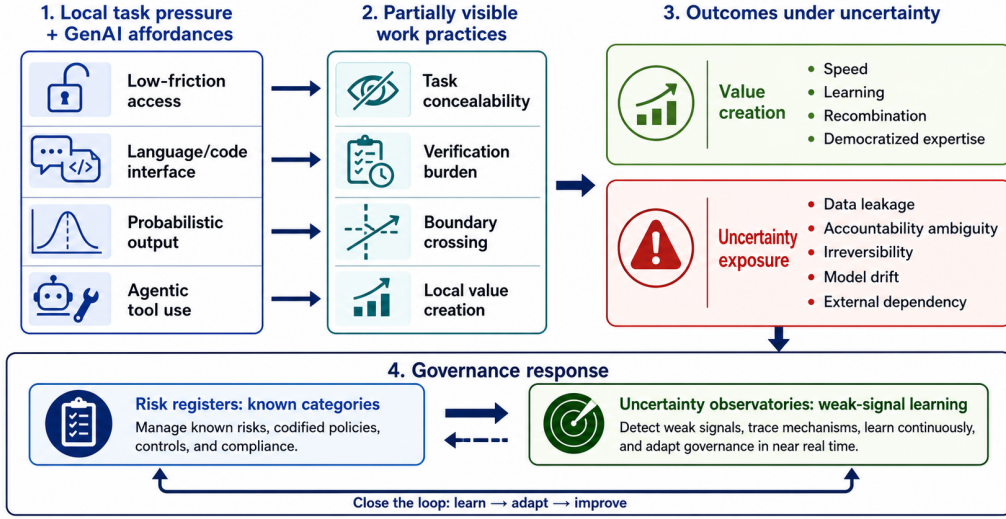


Figure 2: Conceptual problem architecture of shadow GenAI innovation. The figure moves from local task pressure and GenAI affordances to partially visible work practices, governance category lag, and the proposed observatory response.

2. Theoretical background

2.1. Risk, uncertainty, and digital innovation

The distinction between risk and uncertainty is foundational for this paper. In Knightian terms, risk involves outcomes whose categories and probabilities can be estimated; uncertainty involves situations in which the relevant categories, causal relations, or probability distributions are not yet stable (Knight, 1921). Organizations often treat emerging technologies as if they were primarily risk problems. They identify hazards, assess likelihood and impact, assign owners, and define controls. This is useful when the hazard category is known: confidentiality breach, biased decision, hallucinated output, copyright exposure, vendor lock-in, or cybersecurity vulnerability. Yet digital innovation repeatedly produces situations in which the categories themselves

are in motion. Technologies are recombined, users improvise, infrastructures evolve, and organizational routines adjust through use (Yoo et al., 2010; Nambisan et al., 2017). In these conditions, governance requires not only ex ante risk classification but also the capacity to notice weak signals and revise categories as practice unfolds (Ansoff, 1975; Maitlis and Christianson, 2014).

GenAI intensifies this distinction because it changes both the speed and texture of digital experimentation. A conventional enterprise system usually has a visible boundary: it is procured, configured, permissioned, and integrated. A GenAI workflow may be distributed across a prompt, a browser tab, a plugin, a local script, a cloud model, a code editor extension, and a human review step. The object of governance is no longer only a system but a situated task arrangement. A risk register can list “use of external AI tool” as a risk, but the actual uncertainty may concern what employees ask the tool to do, how output is validated, whether the model changes between uses, whether an agent calls another service, whether generated code is copied into production, and whether the organization can reconstruct what happened after an incident.

The problem can be stated as a category-lag problem. A risk category is a stabilized managerial object: it has a name, a boundary, a control owner, and a repertoire of responses. GenAI work practices often precede that stabilization. Employees discover uses at the task edge, vendors alter product capabilities, and models acquire new modalities or integrations. As a result, the organization may have a policy category such as “do not enter confidential information into public AI systems” while the live practice concerns a more complex chain: summarizing a confidential transcript with an enterprise model, copying an excerpt into a public tool for formatting, asking an agent to draft a follow-up email, and storing the output in a customer relationship system. The uncertainty lies in the chain, not only in any single tool. This is why governance must attend to sequences, boundaries, and dependencies.

Digital innovation theory helps explain why category lag is not accidental. Digital artifacts are editable, reprogrammable, and recombining; they can be layered into new services without the full redesign of underlying infrastructure (Yoo et al., 2010; Nambisan et al., 2017). GenAI adds a cognitive and linguistic layer to this generativity. Users do not need to build a full application to innovate. They can create a task-specific micro-routine through prompts, generated code, copied outputs, retrieval, or agent instructions. The cost of experimentation falls, but so does the visibility of experimentation. This produces a governance inversion: the easier a technology is to use for

local innovation, the harder it may be for the organization to know which innovations are occurring.

2.2. From shadow IT and shadow AI to shadow GenAI

Shadow IT research shows that unofficial technology use is not simply deviance. Shadow systems can compensate for slow enterprise systems, local misfit, unmet information needs, and user innovation (Behrens, 2009; Furstenau et al., 2017; Silic and Back, 2014; Kopper et al., 2020). These systems create risks: data fragmentation, security exposure, process inconsistency, and maintenance dependence on local experts. They also create value by allowing users to adapt technology to local tasks. The managerial challenge is therefore ambidextrous: preserve local innovation while reducing uncontrolled exposure.

Chin et al. (2025) provide a direct shadow-AI predecessor by analyzing employees’ shadow AI usage in metaverse-based business models. Their Yin–Yang framing is important because it treats shadow AI as paradoxical: the same behavior can support productivity and innovation while also increasing organizational knowledge leakage and threatening sustainability. Waters-Lynch et al. (2025) develop a second adjacent concept, shadow user innovation, in which employees covertly use consumer-grade GenAI to create value while remaining outside formal oversight. Their model emphasizes concealability, sanction expectations, disclosure incentives, and dynamic-capability renewal. Together, these studies prevent the present article from claiming novelty at the level of identifying shadow AI or covert GenAI use. The open theoretical problem is different: how to govern task chains whose authorization, visibility, data boundaries, output reliance, reversibility, and agentic autonomy vary across steps.

The present paper therefore re-specifies the governance problem. Many consequential GenAI practices are not simply hidden artifacts waiting to be disclosed. They are task chains that may mix approved and unapproved tools, visible and invisible steps, human and agentic action, low-risk drafting and high-stakes reliance. A worker may use an enterprise model for one step, a public model for another, a browser agent for search, a code assistant for modification, and human review for final acceptance. In such cases, the governance question is not only whether an employee violated policy or whether a covert innovation should be revealed. It is whether the organization can interpret a moving chain of use before the chain becomes a dependency,

exposure, or incident. The governance problem is therefore uncertainty regulation across task boundaries. Table 3 makes this positioning explicit.

Table 3: Positioning shadow GenAI task-chain uncertainty against closest predecessor streams

Dimension	Shadow AI usage	Shadow user innovation	Shadow GenAI task-chain uncertainty
Core object	Employees’ unsanctioned AI use and its consequences for organizational knowledge leakage and business-model sustainability (Chin et al., 2025)	Covert employee GenAI innovation outside formal oversight, with potential disclosure and integration into dynamic capabilities (Waters-Lynch et al., 2025)	Partly visible AI-mediated task chains that may combine approved tools, unapproved tools, human verification, data movement, and agentic action
Primary tension	Paradoxical benefit and harm: shadow AI can support performance while increasing knowledge leakage and sustainability risks	Concealment versus revelation: hidden innovation may be sanctioned or converted into organizational renewal	Category lag: task practices change faster than authorization, visibility, verification, and accountability categories stabilize
Governance target	Shadow AI behavior and knowledge-oriented management controls	Disclosure incentives, concealability, and integration of useful covert innovation	Task-level boundaries: data sensitivity, output reliance, reversibility, agent permissions, and downstream action
Evidence role	Direct scholarly predecessor showing that shadow AI is already an empirical governance issue	Closest GenAI-specific innovation predecessor showing that covert employee use can create value	Public proxy evidence map plus theoretical propositions; not a prevalence estimate for hidden conduct
Added leverage here	Keeps knowledge leakage and sustainability costs in view	Keeps employee agency and value creation in view	Explains why even approved tools can become shadow-like when task chains are weakly visible or poorly accountable

The comparison positions the article as an extension of, not a replacement for, adjacent shadow AI and shadow user innovation work. The distinct contribution is the task-chain and uncertainty-governance account.

Shadow GenAI inherits the ambivalence of shadow IT and shadow user innovation but differs in important ways. First, the boundary of the artifact is less stable. A spreadsheet can usually be located. A GenAI practice may be a sequence of prompts and copied outputs across several services. Second, the output is generative and probabilistic. It can appear fluent while being wrong, incomplete, biased, insecure, or unsupported by evidence. Third, the practice may involve latent data transfer: a prompt can contain confidential context, code, customer information, or strategic plans. Fourth, the practice can be agentic. AI agents do not merely suggest; they can search, call tools, write files, create tickets, modify code, or trigger external workflows.

Fifth, the practice is socially normalized by consumer access and professional communities. Employees may view GenAI as a general productivity tool rather than as an enterprise system requiring approval.

These differences mean that shadow GenAI should not be defined only by authorization status. A binary distinction between approved and unapproved tools misses the more important question of visibility and accountability. An approved enterprise copilot can still be used in an ungoverned way if employees rely on output for a high-stakes decision without verification. Conversely, an unapproved model used to brainstorm low-sensitivity wording may create little organizational exposure. We therefore define shadow GenAI innovation along seven dimensions: authorization, visibility, task criticality, data sensitivity, output reliance, reversibility, and agentic autonomy. These dimensions are derived by combining shadow IT and shadow user innovation concerns with authorization and concealment; responsible AI concerns with visibility, auditability, data sensitivity, and accountability; human-AI collaboration concerns with output reliance and verification burden; dynamic capability and resilience concerns with reversibility and learning; and agentic AI concerns with autonomy and boundary crossing. The same tool can be low-shadow in one task and high-shadow in another. This task-level definition aligns with digital innovation theory because it treats innovation as situated recombination rather than as the adoption of a bounded artifact. Table 4 translates the definition into diagnostic dimensions and corresponding observatory functions.

Table 4: Seven dimensions of shadow GenAI task-chain uncertainty

Dimension	Theoretical origin	Public proxy in this study	Observatory function	Governance response
Authorization	Shadow IT and shadow user innovation	Not directly observed; inferred as an inference limit across public data	Separate tool approval from task approval	Safe exception channels; approved/prohibited pattern catalogue
Visibility	Responsible AI auditability and weak-signal sensing	Observability/security tooling among agent users; incident metadata opacity	Detect whether use can be reconstructed	Lightweight disclosure; logging proportional to consequence
Task criticality	Work-practice governance and risk triage	BTOS function differences; Anthropic task/occupation exposure	Prioritize high-consequence work domains	Escalated review for customer, code, legal, safety, financial, or security uses
Data sensitivity	AI risk management and privacy governance	Source-specific limits; incident access and integration signals	Identify prompt, context, and output data boundaries	Data minimization; vendor review; restricted data classes
Output reliance	Automation-augmentation and situated trust	Stack Overflow trust, complex-task performance, and use frequency	Distinguish draft support from decision, code, or action	Verification protocols; human review thresholds
Reversibility	Incident response and dynamic capabilities	Incident duration, partial degradation, and operational workarounds	Assess whether consequences can be recalled or rolled back	Rollback plans; containment playbooks; versioning
Agentic autonomy	Agent governance and boundary accountability	Agent use, orchestration, external tools, and observability/security stack	Map actions across systems and owners	Permission scoping; human approval points; boundary maps

The dimensions are not a validated psychometric scale. They are a theoretical diagnostic for locating where AI-mediated work becomes more shadow-like as authorization, visibility, reliance, reversibility, and autonomy shift.

The purpose of these dimensions is diagnostic and design-oriented, not psychometric validation.

2.3. Responsible AI governance and work-practice blind spots

Responsible AI governance has developed important instruments: model documentation, impact assessment, internal auditing, risk management frameworks, accountability processes, and ethical principles (Mitchell et al., 2019; Raji et al., 2020; AI, 2023; Floridi and Cows, 2022; Jobin et al., 2019). These instruments matter. They bring structure to model development and deployment, create accountability artifacts, and help organizations document due diligence. However, many were designed for visible systems. Model cards presume a model that can be documented. Algorithmic audits presume an

object of audit. AI risk management frameworks presume that organizational actors can identify, map, measure, and manage the system under review. Shadow GenAI strains these assumptions because the governance object may be a work practice that emerges before it is registered as a system.

The blind spot is not a failure of responsible AI as a field. It is a scale and visibility problem. Organizational AI increasingly appears in learning algorithms embedded in work and in algorithmic control within contested organizational terrain (Faraj et al., 2018; Kellogg et al., 2020). GenAI broadens this by making algorithmic capability available through ordinary language interfaces. The challenge is not only how to govern formal AI systems but also how to govern AI-mediated work when the line between tool, task, and user practice is blurred. This is especially true for software development, where AI assistants and agents can move rapidly from suggestion to code change, test generation, debugging, documentation, and deployment support, while inherited lessons from machine-learning engineering show how invisible dependencies and technical debt can accumulate when AI-mediated systems are weakly documented (Sculley et al., 2015; Amershi et al., 2019).

Prior research on AI and management emphasizes a familiar automation-augmentation tension: AI can substitute for human work while also augmenting judgment, reshaping decision structures, and requiring new forms of human-machine collaboration (Raisch and Krakowski, 2021; Shrestha et al., 2019; Berente et al., 2021). Shadow GenAI adds a second tension. The more locally useful a GenAI tool becomes, the more readily it can be adopted before formal governance understands the task, data, and accountability conditions around it. Value creation and governance exposure are therefore coupled. Policy that focuses only on preventing misuse can miss learning opportunities; policy that celebrates experimentation without observability can create avoidable harm.

This paradox is visible in the difference between model governance and work-practice governance. Model governance asks whether a model is documented, tested, monitored, and deployed responsibly. Work-practice governance asks whether people use model outputs in ways that are visible, verified, accountable, and appropriate to task context. Both are needed. A well-documented model can still be embedded in a poorly governed workflow, and a risky model can sometimes be used safely in a low-stakes exploratory task. The governance target is therefore a socio-technical configuration: model, interface, user, task, data, output, downstream action, and organizational response.

2.4. *Uncertainty observatories as dynamic capability*

Dynamic capabilities theory explains why organizations cannot solve the problem through prohibition alone. Dynamic capabilities involve sensing, seizing, and transforming under changing environments (Teece, 2007; Teece et al., 2016). Under high-velocity conditions, capabilities may also be simple, experiential, and improvisational rather than fully specified routines (Eisenhardt and Martin, 2017). GenAI creates a similar tension. Organizations need to sense emergent uses, seize value from experimentation, and transform routines for responsible deployment. But the sensing problem is difficult because the most informative activity may occur at the edge of formal visibility. Employees learn which prompts work, which tools fail, where output is nearly correct, and when policy feels too slow for operational demands. Those local discoveries can be valuable signals. If governance treats them only as violations, it may suppress the information needed to update policy.

We use the term *uncertainty observatory* to describe an organizational capability for detecting, interpreting, and learning from weak signals of emerging GenAI use. The term builds on weak-signal strategy (Ansoff, 1975), sensemaking (Maitlis and Christianson, 2014), exploration-exploitation learning (March, 1991), dynamic capabilities (Teece, 2007), and AI accountability research (Raji et al., 2020). An observatory is not a dashboard, an AI center of excellence, or a surveillance program. It is a coupled set of practices: safe task-level disclosure, model and agent inventory, near-miss reporting, incident review, boundary mapping, human verification protocols, reversibility checks, and policy learning. Its purpose is to complement risk registers by capturing uncertainty that cannot yet be reduced to stable categories.

This framing connects digital capability gaps to value slippage. When formal tools are unavailable or too slow, useful experimentation may be suppressed, creating value missing. When employees work around governance without visibility, useful experimentation can become value slippage: local gains are offset by hidden dependencies, data exposure, verification burden, or institutional mistrust. Uncertainty observatories are therefore a form of uncertainty regulation. They do not eliminate uncertainty; they create legitimate channels through which uncertain practice can be noticed, interpreted, and gradually converted into governable risk knowledge.

2.5. *Propositions*

The preceding argument can be summarized as seven propositions that organize the empirical studies and the governance framework. These propo-

sitions are theoretical claims for future organizational testing rather than statistical hypotheses tested directly in the public-data design.

Proposition 1: Diffusion pressure. The greater the availability and work relevance of GenAI tools, the more likely employees are to initiate AI-mediated task experimentation outside formal innovation portfolios, especially when approved tools lag behind task demands.

Proposition 2: Category lag. Shadow GenAI pressure increases when work practices evolve faster than organizational categories for approved tools, data classes, verification requirements, and accountable ownership.

Proposition 3: Verification burden. Mixed trust in GenAI outputs does not eliminate use; instead, it shifts burden to local verification routines, making output reliance and human review central dimensions of governance.

Proposition 4: Agentic boundary crossing. As GenAI moves from content generation to agentic action, governance exposure shifts from output quality alone to permissions, tool calls, downstream systems, and reversibility.

Proposition 5: Constraint shadowing. Policies that are slow, punitive, or mismatched to work needs can increase concealment, whereas safe and rapid disclosure channels can convert hidden practice into institutional learning.

Proposition 6: Weak-signal learning. Organizations that collect and interpret near misses, provider degradation, workarounds, and verification failures will update risk categories faster than organizations that rely only on static risk registers.

Proposition 7: Legitimate observability. Observability improves governance only when it is proportionate, purpose-limited, contestable, and trusted; otherwise, it can intensify surveillance concerns and drive shadow practices further underground.

3. Research design

The empirical design is a mechanism-oriented public evidence map. It does not claim to observe every instance of shadow GenAI directly. Such a claim would be inappropriate because shadow use is partly hidden, and public datasets rarely contain verified measures of organizational authorization.

Instead, the design asks whether multiple public evidence streams identify conditions that make shadow GenAI innovation plausible and governance-relevant: widespread adoption pressure, task-level AI use, mixed trust and verification burden, agentic tooling, uneven governance instrumentation, and operational weak signals that do not fit stable risk categories. This is a mechanism-tracing logic rather than a single-dataset causal identification design.

The design has three studies. Study 1 is an author-conducted structured synthesis of the literature. It maps how GenAI, responsible AI, organizational work, digital innovation, uncertainty, and shadow IT are currently connected, and it adds targeted closest-predecessor checks on shadow AI usage and shadow user innovation. Study 2 analyzes public adoption and task-exposure data: the U.S. Census BTOS AI supplement, Stack Overflow Developer Survey 2025, and Anthropic Economic Index. Study 3 codes public AI provider-status incidents as weak signals of operational uncertainty. The studies are then integrated into a framework for uncertainty observatories and the propositions developed in Section 2. Table 5 summarizes what each evidence stream can and cannot support.

Table 5: Evidence streams, mechanism signals, and inference boundaries

Evidence stream	Mechanism-oriented signal	What it does not support	Role in the mechanism argument
Literature synthesis and predecessor reading	Conceptual concentration around GenAI, governance, risk, organizational work, plus targeted positioning against shadow AI usage and shadow user innovation	Prevalence claims or proof that the entire adjacent full-text literature has been exhausted	Shows category fragmentation, prevents novelty overclaiming, and motivates theory-building
BTOS AI supplement	Representative firm-level AI diffusion by function, sector, size, and geography	Individual employee behavior, authorization status, or internal governance routines	Establishes organizational adoption pressure
Stack Overflow 2025	Developer AI use, trust, AI-agent use, friction, and governance-tool signals	All-worker generalization or verified compliance behavior	Shows work-practice normalization, mixed trust, and uneven instrumentation among a technically salient population
Anthropic Economic Index	Occupation and task exposure/penetration pressure	Authorization, hidden use, or automation-versus-augmentation in the local files	Locates where task pressure may concentrate
Provider-status incidents	Operational weak signals such as partial degradation, access failures, latency, and technical opacity	AI harm prevalence, provider reliability rates, or internal organizational workarounds	Shows infrastructural conditions that can trigger improvised GenAI workarounds

The design is triangulated and mechanism-oriented. No single evidence stream is treated as a direct measure of shadow GenAI prevalence.

3.1. Literature synthesis and corpus construction

The authors conducted a structured literature search using OpenAlex and prior OpenAlex/Crossref seed records, with query families covering GenAI, organizational AI governance, responsible AI, shadow IT, digital innovation under uncertainty, AI agents, and AI risk management (OpenAlex, 2026). The pipeline identified 465 records before deduplication, screened 401 deduplicated records, excluded 348 low-relevance records, and retained 53 records for synthesis. Screening was rule-based and transparent: records needed to connect AI/GenAI to organization, work, governance, risk, uncertainty, digital innovation, or agentic systems. Broad matches without organizational relevance were excluded. Topic modeling used a non-negative matrix factorization representation of title, abstract, and concept text. Keyword co-occurrence used controlled terms and document-level co-presence.

This procedure is best described as a structured, search-based literature synthesis with transparent corpus boundaries. It combines bibliographic records, available abstracts, concept fields, topic modeling, keyword co-

occurrence, and targeted closest-predecessor reading. It does not claim to exhaust every full-text article in all adjacent fields, but it is more than a bibliographic count: the synthesis is used to position the paper theoretically and to show why shadow GenAI remains dispersed across governance, responsible AI, digital innovation, shadow IT, and work-practice literatures. If direct shadow GenAI terminology is sparse while governance, GenAI, and organizational-work themes are abundant, the absence is theoretically informative: scholarship has many pieces of the puzzle but has not yet stabilized a vocabulary for partially visible GenAI innovation. The word cloud in Figure 3 previews the corpus vocabulary that motivates this mapping exercise. Because directly relevant 2025 articles on shadow AI usage and shadow user innovation appeared at the edge of this fast-moving literature, the manuscript also uses targeted predecessor checks to position the present contribution without changing the corpus counts.



Figure 3: Word cloud representing the retained 53-record literature corpus. Term sizes reflect frequency after stopwords filtering and light weighting of core conceptual anchors used in the corpus-screening logic.

3.2. Public adoption and exposure data

The BTOS AI supplement provides U.S. employer-business estimates for current AI use, AI use by function, worker AI and GenAI use, expected future use, sector, state, and employment-size classes (U.S. Census Bureau, 2026b,a). The local pipeline downloaded the Census CSV/workbook files from the BTOS visualization assets and parsed percentage estimates into long-form tables. The Stack Overflow Developer Survey 2025 provides item-level responses for 49,191 respondents (Stack Overflow, 2025). The pipeline derived indicators for current AI use, weekly/daily and daily AI use, AI-output trust, favorable sentiment, complex-task performance, Stack Overflow friction caused by AI tools, agent use, orchestration tools, external agents, observability/security tools, and agent-related challenges. It also derived covariates for professional developer status, employment, work experience, role group, country group, organization size, remote work, people-management status, and education. The Anthropic Economic Index provides occupation exposure and task penetration measures (Anthropic, 2026). Because the local Anthropic files do not include a direct augmentation-versus-automation label, the paper treats those measures conservatively as exposure and penetration proxies.

The main Stack Overflow models are logistic regressions and ordered logits. M1 estimates any AI-tool use. M2 estimates positive trust in AI-output accuracy among AI users. M3 estimates AI-agent use. M4 estimates observability/security stack use among agent users. Robustness checks re-estimate the logic with alternative outcome thresholds: weekly/daily AI use, daily AI use, nonnegative trust, weekly/daily agent use, agent planned-or-used, and a broader agent-governance-stack measure. These models are associational. They are used to identify patterns consistent with adoption pressure and governance instrumentation, not to claim causal effects of individual traits.

The binary-outcome models can be summarized with the following template specification:

$$\Pr(Y_i = 1) = \text{logit}^{-1}(\alpha + \beta_1 \text{Exposure}_i + \beta_2 \text{Trust}_i + \beta_3 \text{AgenticTooling}_i + \gamma X_i), \quad (1)$$

where Y_i is the model-specific outcome and X_i contains the available controls for role, country group, work experience, employment status, work location, people-management responsibility, education, and organization size. Equation (1) is a template representation of the binary-outcome models; each reported

model uses a model-specific outcome, analysis sample, and predictor set, as described in Table 10 and Section Appendix C and the variable construction scripts. The exact predictor set differs by model because some predictors are only meaningful after AI or agent use is reported. For example, observability/security tooling is estimated among agent users because non-agent users have no reason to report agent observability tools. Ordered-logit models use the same logic for frequency and trust scales.

Three diagnostic constructs guide interpretation. The *adoption-governance gap* is not a single universal score because the public datasets have different units and denominators. Instead, it is observed when adoption indicators are high while trust, verification, or governance-tool indicators remain materially lower. The *shadow pressure* construct is inferred when high adoption intersects with task criticality, strict policy friction, external tool availability, and low observability. The *uncertainty signal intensity* construct is inferred when incidents or survey responses indicate partial degradation, technical opacity, unknown status, boundary coupling, or verification burden. These constructs discipline the inference: the paper does not equate adoption with shadow use; it treats adoption as one necessary condition that becomes governance-relevant when combined with low visibility and high uncertainty.

Missingness is handled conservatively. The Stack Overflow survey uses item-specific denominators because respondents do not answer every AI or agent item. The manuscript reports denominators where substantively important and avoids comparing percentages as if all came from the same denominator. BTOS estimates are public aggregate percentages rather than microdata. Anthropic task and occupation measures are treated as descriptive exposure indicators. Incident duration is used only where start and resolution timestamps are available. All generated tables and figures are produced from replication scripts and then copied into the manuscript project.

3.3. Incident weak-signal coding

The incident study uses public AI provider-status incidents from the AI Status Dashboard feed (AI Status Dashboard, 2026). The raw feed contains titles, providers, dates, status, severity, and resolution times but not rich narrative descriptions. The pipeline therefore codes weak signals from title/status metadata only. This complements incident-database work that treats AI failures as learnable public records (McGregor, 2021; Responsible AI Collaborative, 2026), but it is narrower and more operationally focused. Categories include API errors/availability, authentication/access,

data pipeline/integration, latency/performance, model quality/behavior, and uncoded operational signals. Uncertainty signals include partial degradation, technical opacity, unknown or investigating status, and external dependency. This source is not a population-level AI harm database. It is useful because provider-status events reveal operational fragility around the infrastructures on which shadow and formal GenAI workflows both depend. Table 6 records the metadata-synthesis search and screening protocol, and Figure 4 reports the corresponding screening flow.

Table 6: Structured literature review protocol and corpus accounting

Stage	Operational rule	Count	Inference role
Identification	OpenAlex enrichment plus prior OpenAlex/Crossref seed search across GenAI, shadow IT, digital innovation, uncertainty, responsible AI, risk, and agentic AI	465	Defines the evidence-map universe
Deduplication	DOI matching plus normalized-title matching	401	Removes repeated records across query families
Screening	Rule-based title/abstract/concept/venue scoring; broad biomedical/general noise removed	348 excluded	Protects the synthesis from false topical matches
Inclusion	Records retained only when AI/GenAI was connected to organization, work, governance, risk, or uncertainty	53	Forms the corpus for topic, keyword, and theme analysis

The review is a transparent, search-based literature synthesis with bounded corpus claims. It is not claimed as a dual-coded exhaustive full-text systematic review.

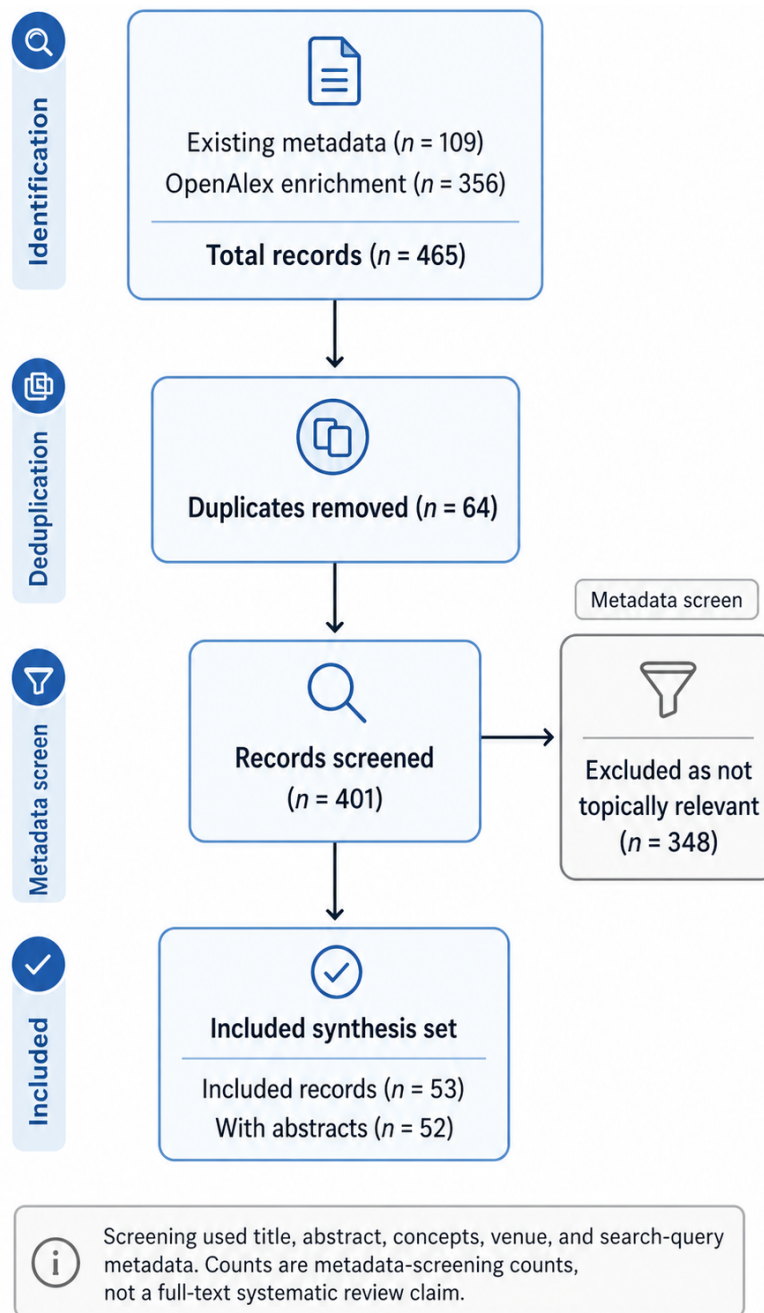


Figure 4: PRISMA-style literature corpus construction and screening flow. The figure reports search-and-screening counts for the structured synthesis rather than exhaustive full-text eligibility counts.

Figure 4 reports the identification, deduplication, exclusion, and inclusion counts for the authors’ structured literature study; it does not make the stronger claim of a dual-coded exhaustive full-text systematic review. In Figure 5, we present our analysis pipeline and inference boundaries that separate corpus construction, public-data cleaning, model estimation, incident coding, and manuscript-facing tables and figures.

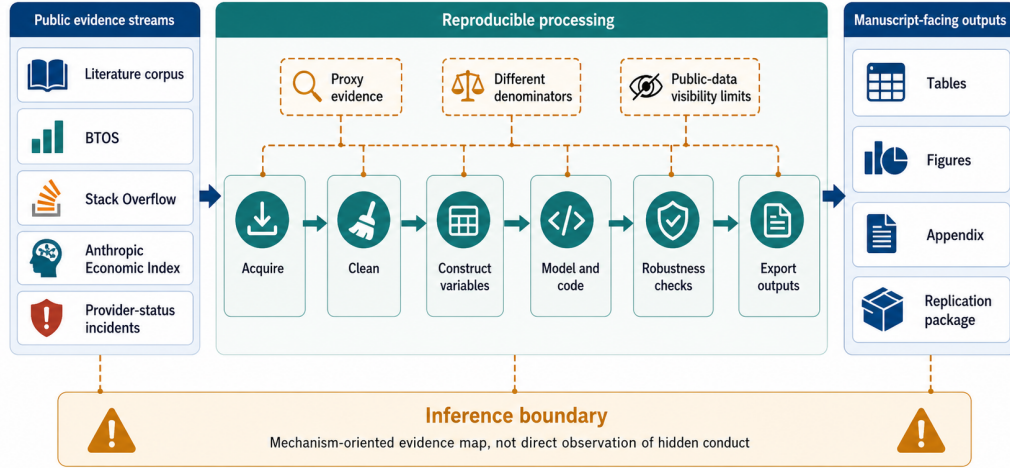


Figure 5: Analysis pipeline and inference boundaries. The design separates corpus construction, public-data processing, model estimation, incident weak-signal coding, and manuscript-facing outputs.

4. Study 1: Literature synthesis

The first study examines whether existing research already provides a mature conceptual language for shadow GenAI innovation. The answer is mixed. The author-conducted structured literature synthesis finds substantial research on governance, risk, responsible AI, GenAI, and organizational work, but little direct consolidation around shadow GenAI as a distinct socio-technical phenomenon. Among the 53 included records, 40 contain governance/risk/responsible-AI signals, 38 contain GenAI signals, 34 contain organization/work signals, and 9 contain agentic-AI signals. No included record carried a direct controlled signal for shadow AI or shadow IT after screening. This absence should not be overread as proof that shadow use

is absent in practice. It indicates that the search-indexed scholarly vocabulary has not yet stabilized around a workplace phenomenon that is salient in contemporary organizational debate but still dispersed across separate literatures.

The topic model reinforces this interpretation. The largest metadata-derived topic emphasizes ethical, privacy, sustainable, and governance language. Other topics concentrate on language-model capabilities and tasks, education and learning, management and supply-chain integration, and health-care applications. These topics show that GenAI scholarship has expanded quickly, but much of it remains organized around sectoral application domains or normative governance principles rather than around partially visible workplace innovation. The management and supply-chain topic is closest to the organizational innovation tradition; the language-model topic is closest to technical capability; the ethics topic is closest to responsible AI. The gap lies between them: how employees and teams convert general-purpose GenAI capability into situated work practices that may not be known to formal governance actors.

The targeted closest-predecessor check sharpens the point. Chin et al. (2025) study shadow AI usage in metaverse-based business models and show that shadow AI can have conflicting effects on organizational knowledge leakage and sustainability. That article establishes shadow AI as a direct empirical governance problem and makes clear that the present paper should not frame shadow AI itself as newly discovered. Waters-Lynch et al. (2025) define shadow user innovation as covert GenAI-enabled value creation and theorize disclosure incentives, concealability, and dynamic-capability renewal. That article solves another important part of the puzzle: it explains why hidden GenAI tinkering may be valuable and why organizations should not treat every covert practice as pure deviance. The present article extends the problem in three ways. First, it treats shadow GenAI as a gradient of authorization, visibility, reliance, reversibility, and autonomy rather than only as unsanctioned AI use or covert user innovation. Second, it places the governance problem in VUCA conditions where categories form through weak signals and institutional learning. Third, it uses public empirical proxies to ground the adoption-governance gap while explicitly stating what those proxies cannot prove.

This gap matters because shadow GenAI is not adequately captured by either celebratory innovation narratives or compliance-centered risk narratives. Digital innovation research explains why users recombine technology and why

innovation can emerge outside formal project structures (Yoo et al., 2010; Nambisan et al., 2017; Von Hippel, 2006). Responsible AI research explains why organizations need documentation, accountability, and risk management (Mitchell et al., 2019; Raji et al., 2020; AI, 2023). Shadow IT and shadow user innovation research explain why users adopt or create solutions outside formal control (Behrens, 2009; Furstenuau et al., 2017; Waters-Lynch et al., 2025). But GenAI requires a synthesis because the innovation object is not necessarily a system. It may be a prompt chain, a generated artifact, an agentic action, a verification routine, or a workaround that moves across several tools. Figure 6 reports the literature diagnostics underlying this interpretation.

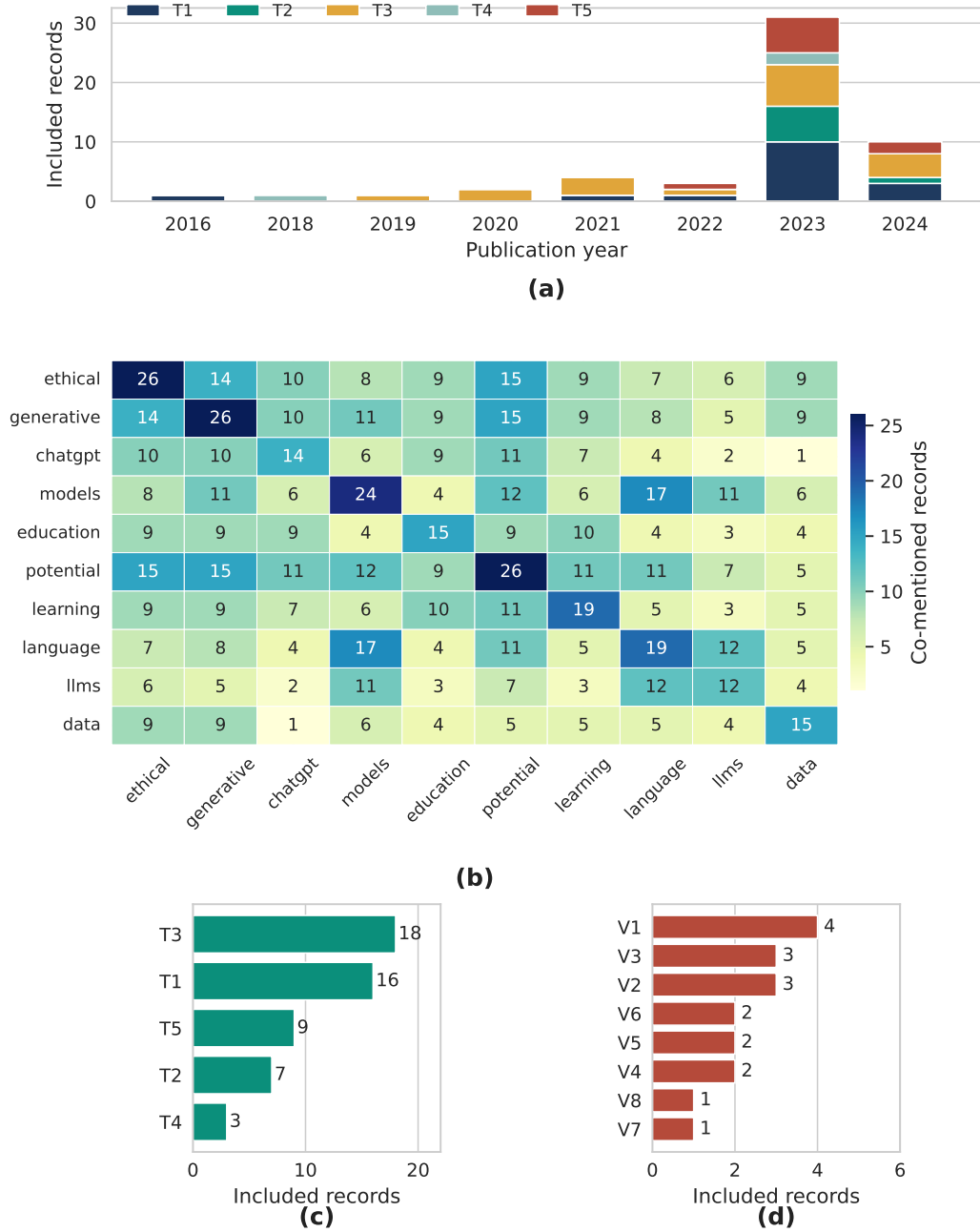


Figure 6: Literature synthesis diagnostics: (a) publication-year distribution by compact topic cluster; (b) keyword co-occurrence intensity among the highest-frequency terms; (c) topic concentration using compact topic IDs; and (d) venue concentration using compact venue IDs. Topic and venue keys are reported in Tables 7 and 8.

Tables 7 and 8 define the compact topic and venue IDs used in panels (c) and (d).

Table 7: Topic key for Figure 6(c)

Topic ID	Interpretable keyword cluster	Included records
T3	ethical; gai; ethics; sustainable	18
T1	llms; language; humans; tasks	16
T5	health; care; health care; healthcare	9
T2	education; higher education; higher; genai	7
T4	management; supply chain; chain; supply	3

Topic IDs are compact labels used only to keep the square panel readable.

Table 8: Venue key for Figure 6(d)

Venue ID	Venue	Included records
V1	Unspecified venue	4
V2	IEEE Access	3
V3	arXiv (Cornell University)	3
V4	International Journal of Educational Technology in Higher Education	2
V5	International Journal of Information Management	2
V6	Artificial Intelligence Review	2
V7	Computational Linguistics	1
V8	Journal of Applied Learning & Teaching	1

Venue IDs are compact labels used only to keep the square panel readable.

The panels in Figure 6 should be read as diagnostics rather than as a claim of exhaustive literature settlement. Panel (a) shows the recency of the included corpus and the rapid concentration of GenAI-governance work after the diffusion of large language models. Panel (b) shows that high-frequency terms connect governance, language models, ethics, learning, and management, but do not form a dense shadow-use cluster. Panel (c) shows the dominance of ethics/governance and language-model topics relative to smaller management and supply-chain clusters. Panel (d) shows venue dispersion, which is expected for an emerging cross-disciplinary problem. These diagnostics support the paper’s claim of conceptual fragmentation.

Three theoretical implications follow from Study 1. First, shadow GenAI should be treated as a boundary concept linking digital innovation, shadow IT, shadow user innovation, and responsible AI. It names a phenomenon that is not reducible to any one literature. Second, uncertainty should be centered rather than treated as a residual category. The literature contains many known-risk categories, but the emerging problem is how new categories become visible.

Third, governance should be framed as learning. If organizations only ask whether a tool is approved, they miss how work practices evolve; if they only ask whether a model is risky, they miss where employees are already using it to create value.

The absence of a direct shadow GenAI cluster is analytically useful because it prevents the paper from treating the construct as already settled. Shadow IT studies provide a vocabulary of unauthorized or business-managed systems, but they do not fully address probabilistic output, prompt-mediated work, foundation-model updates, or agentic action. Responsible AI studies provide a vocabulary of documentation, accountability, and risk management, but they often presume a visible model or system. Digital innovation studies provide a vocabulary of generativity and distributed recombination, but they do not always specify how governance learns from hidden or semi-hidden work. Shadow AI usage research provides a vocabulary of unsanctioned AI behavior, knowledge leakage, and paradoxical harm/benefit. Shadow user innovation provides a vocabulary of covert GenAI experimentation and revelation incentives. The task-chain view is needed when official and unofficial elements are entangled, when the same tool can be low- or high-shadow depending on the task, and when agentic workflows cross organizational boundaries. The proposed construct therefore works as a bridge.

This interpretation also clarifies the role of responsible AI principles. Principles such as transparency, accountability, fairness, privacy, and robustness remain important (Floridi and Cowls, 2022; Jobin et al., 2019). The difficulty is operationalizing them when the relevant practice is not a stable AI product. For example, transparency may require not only model documentation but disclosure that a work product was AI-assisted. Accountability may require not only a model owner but a task owner who verified output. Robustness may require not only benchmark testing but monitoring of workflow failure, near misses, and model behavior changes. Privacy may require not only vendor review but prompt-level and data-context awareness. These operational translations are precisely what uncertainty observatories are meant to support.

Study 1 informs RQ1 by showing why shadow GenAI should be theorized as task-chain uncertainty rather than merely unauthorized tool use. It also informs RQ3 by identifying the governance blind spot that the observatory framework must address: scholarship has strong tools for responsible AI systems and for shadow IT, but weaker tools for learning from partially visible AI-mediated work practices.

5. Study 2: Public evidence of the adoption-governance gap

Study 2 examines whether public datasets show conditions consistent with an adoption-governance gap. The evidence is consistent and useful as proxy evidence, but it remains indirect rather than a direct observation of unauthorized conduct. The BTOS AI supplement shows that AI use has diffused into U.S. employer businesses at a scale that makes informal experimentation plausible. Nationally, 17.9% of businesses report current AI use in any business function. Adoption is uneven: 51 - Information reports 37.6%, and the largest employment-size class reports 30.6%. The most common national AI function is Sales and marketing at 14.3%. These figures matter because shadow GenAI pressure is not evenly distributed. It may concentrate where AI is already normalized, where knowledge work is intensive, where firm size increases coordination complexity, and where business functions face time pressure to produce outputs. The pattern sits within a wider shift in which GenAI is moving from experimental productivity support into routine knowledge work (Kshetri et al., 2023; Noy and Zhang, 2023; Stanford Institute for Human-Centered Artificial Intelligence, 2026).

The Stack Overflow data show the worker-side version of the same pattern. Among valid respondents, 78.5% report current AI-tool use, 64.8% report weekly or daily use, and 47.1% report daily use. AI is therefore not an occasional novelty for many developers; it is part of routine work. At the same time, trust is not settled. Only 32.8% report positive trust in AI-output accuracy, while 54.3% are neutral or better. This split is theoretically important. Shadow GenAI is not driven only by naive overtrust. It can also be driven by pragmatic use under uncertainty: workers use tools because they are useful enough, but then must spend effort verifying, debugging, re-prompting, or deciding when human expertise is required. Table 9 summarizes the public-data indicators behind these claims.

Table 9: Descriptive evidence panel for the adoption-governance gap

Evidence stream	Measure	Value	Interpretation
Panel A. Firm adoption pressure			
BTOS	Businesses using AI in any function in the last two weeks	17.9%	Firm-level diffusion is no longer confined to a few experimental organizations
	Highest-adoption sector	51 - Information (37.6%)	Knowledge-intensive sectors have stronger adoption pressure
	Most common AI business function	Sales and marketing (14.3%)	AI enters market-facing and strategic work, not only back-office automation
	Largest employment-size estimate	30.6%	Larger firms face broader coordination and governance exposure
Panel B. Developer practice and governance instrumentation			
Stack Overflow	Current AI-tool use among valid respondents	78.5%	Developer work is already AI-mediated
	Weekly/daily AI use	64.8%	AI is habitual for many respondents, increasing normalization pressure
	Positive AI-output trust	32.8%	Trust is mixed, so value creation remains tied to verification burden
	AI-agent use	30.9%	Agentic work is visible but still unevenly instrumented
	Agent users with observability/security tools	27.3%	Use and governance instrumentation do not move in lockstep
Panel C. Task exposure and operational weak signals			
Anthropic	Mean occupation observed exposure	7.7%	Exposure is concentrated rather than evenly distributed
Incidents	Top failure category	api errors availability (16 of 25)	Operational weak signals cluster around reliability and boundary fragility

Percentages use the valid denominator for the item. Incident counts describe the public status feed used in this analysis.

The model results provide a more technical picture. M1 indicates that professional developers (OR=1.38), people managers (OR=1.48), and remote/hybrid workers (OR=1.20) have higher odds of reporting AI use, net of role and country group controls. M2 shows that positive trust in AI-output accuracy is strongly associated with daily AI use (OR=2.84) and especially with believing AI handles complex tasks well (OR=7.24). This result does not show that use causes trust or trust causes use. It does show that trust and perceived task capability are tightly coupled in this developer population. In governance terms, the relevant object is not simply access to AI but the relationship between repeated use, perceived competence, and verification

effort. Table 10 reports the model estimates.

M3 shows that daily AI use is strongly associated with AI-agent use (OR=3.45), while people managers also have higher odds of reporting agent use (OR=1.42). This suggests that agentic work is not detached from everyday AI adoption; it grows from the same normalization process. M4 then shows the governance-instrumentation layer: among agent users, orchestration tools strongly predict observability/security stack use (OR=4.67), and external agent/copilot use also predicts such tooling (OR=4.78). Remote/hybrid work is associated with observability/security stack use among agent users (OR=1.44). The interpretation is nuanced. Some technically mature users do connect agent use to governance tooling. But the descriptive denominator remains sobering: only 27.3% of AI-agent users report observability or security tools.

Table 10: Selected logit model estimates for AI use, trust, agent adoption, and governance tooling

Model	Predictor	OR	95% CI	N	Pseudo R2
Panel A. AI-use normalization					
M1 AI use any	Professional developer	1.38***	[1.24, 1.52]	25,266	0.045
	People manager	1.48***	[1.32, 1.67]	25,266	0.045
	Remote/hybrid work	1.20***	[1.10, 1.31]	25,266	0.045
Panel B. Output trust and verification burden					
M2 positive AI-output trust	Daily AI use	2.84***	[2.65, 3.05]	22,897	0.230
	AI handles complex tasks well	7.24***	[6.75, 7.77]	22,897	0.230
	AI frustration count	0.95**	[0.92, 0.98]	22,897	0.230
Panel C. Agent adoption and instrumentation					
M3 agent use any	Daily AI use	3.45***	[3.23, 3.67]	23,967	0.100
	People manager	1.42***	[1.29, 1.56]	23,967	0.100
M4 agent observability/security stack among agent users	Uses agent orchestration tools	4.67***	[4.15, 5.26]	6,552	0.136
	Uses external agents/copilots	4.78***	[3.34, 6.86]	6,552	0.136
	Remote/hybrid work	1.44***	[1.22, 1.71]	6,552	0.136

Models use Stack Overflow 2025 public responses. Stars denote $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***). Estimates are associational and item-denominator specific.

Figure 7 visualizes the adoption-to-governance funnel, selected odds ratios, BTOS sector-function adoption, and Anthropic exposure distribution.

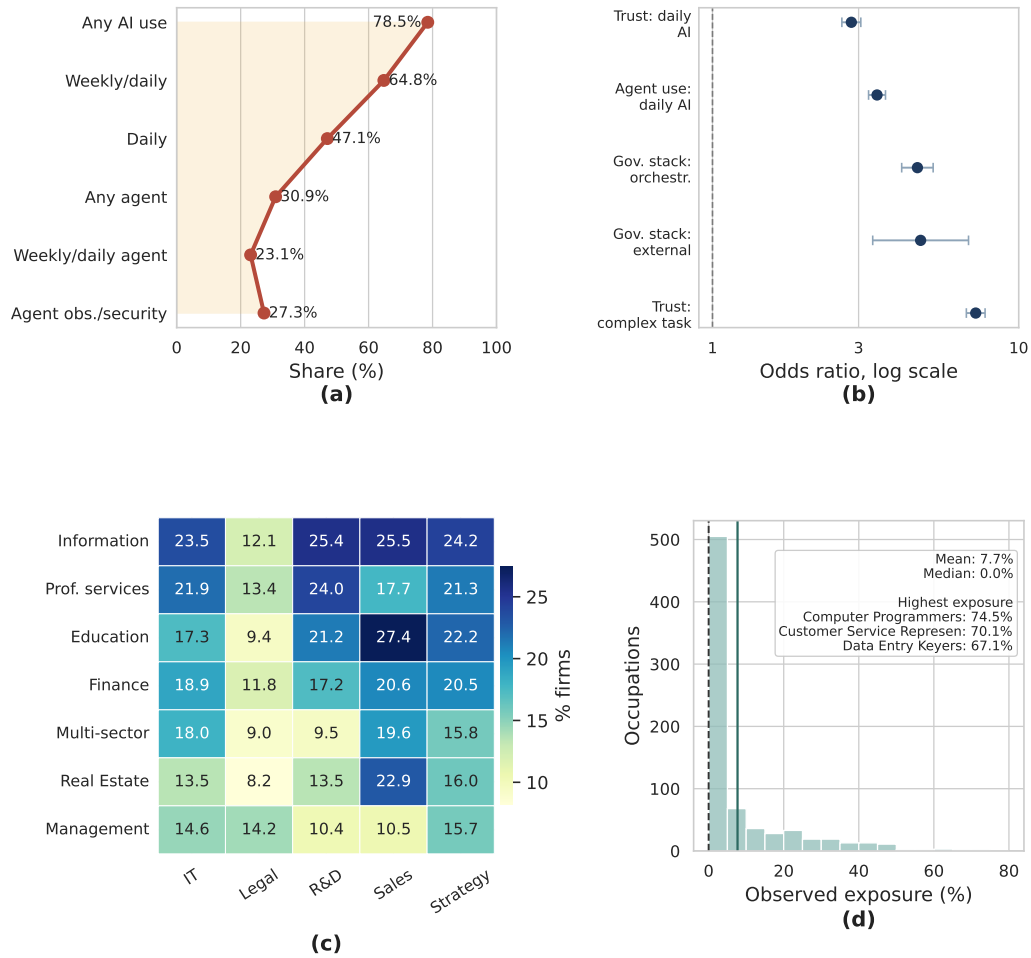


Figure 7: Public evidence dashboard for the adoption-governance gap: (a) Stack Overflow adoption-to-governance funnel; (b) selected Stack Overflow odds ratios for trust, agent use, and governance instrumentation; (c) BTOS sector-function adoption heatmap; and (d) Anthropic occupation exposure distribution.

The Anthropic data add task and occupation texture. The mean occupation observed exposure is 7.7%, but exposure is highly concentrated. The highest-exposure occupation in the local file is Computer Programmers at 74.5%. This concentration is important because GenAI governance is not only an enterprise IT problem. It is a work-redesign problem. Occupations and tasks with high exposure face stronger incentives to integrate AI into everyday routines, which can generate both productivity gains and new dependencies.

Where exposure is concentrated, governance should be close to work practice rather than centralized only in abstract policy.

Robustness checks support the direction of the main findings. The pipeline re-estimated the models with alternative thresholds for AI-use frequency, trust, agent use, planned-or-current agent adoption, and broad governance-stack measures. Across these specifications, adoption intensity, perceived task capability, and agentic tooling remain systematically related to trust, agent use, and governance instrumentation. The specific coefficients vary by threshold, as expected, but the substantive inference does not depend on a single binary cut point.

Study 2 is consistent with four bounded claims. First, adoption pressure is high enough that GenAI experimentation outside formal innovation portfolios becomes plausible, especially where approved tools and task needs are misaligned. Second, trust is mixed rather than uniformly high, so shadow GenAI governance must address verification burden rather than only preventing reckless use. Third, agentic use is a meaningful extension of routine AI use in the developer population, not merely a distant future scenario. Fourth, governance instrumentation is uneven, which is precisely the condition under which an uncertainty observatory becomes valuable.

The technical depth of these findings lies in the relationship among the descriptive and model-based signals. The BTOS estimates establish that organizational AI use is not restricted to technology firms or isolated pilots. Sector and size gradients indicate that diffusion is structured by economic activity and organizational scale. The Stack Overflow results then show that, within a technically sophisticated worker population, AI use is frequent but epistemically unsettled. The Anthropic data show that exposure is concentrated in occupations and tasks where language, code, customer interaction, and information processing are central. These three layers together describe a plausible pathway: task exposure creates incentives for individual experimentation; repeated use normalizes AI-mediated work; mixed trust creates local verification routines; and agentic tools increase the need for observability, security, and boundary mapping.

The adoption-governance gap is therefore not simply a difference between one high percentage and one low percentage. It is a patterned misalignment across layers. Firm-level adoption indicates organizational exposure. Individual-level use indicates work-practice normalization. Trust and friction indicators reveal that output evaluation remains unsettled. Agent-use indicators show movement from content generation toward delegated action.

Governance-tool indicators show that instrumentation exists but is not universal. A risk register can record each of these items separately, but an uncertainty observatory is needed to connect them as a changing system of work.

The Stack Overflow trust model is especially revealing. Positive trust is not widespread, yet it is strongly associated with perceived complex-task performance. This suggests that respondents may not trust AI in the abstract; they trust it when they experience it as competent for the tasks they care about. That form of situated trust can be productive, but it can also be risky. If a tool performs well in routine cases, employees may extend reliance to edge cases where verification is harder. Governance must therefore avoid both cynicism and overconfidence. It should ask which tasks produce justified confidence, which tasks produce misleading fluency, and which tasks require mandatory human review.

The agent results extend this point. Daily AI use predicts agent use, which means agentic workflows can grow out of familiar AI-assistant practices rather than arrive as a wholly separate technology category. This has practical implications for timing. Organizations that wait to govern agents until after they have a formal agent strategy may already have agent-like practices in code editors, workflow tools, browser assistants, and productivity suites. The strong association between orchestration tools and observability/security tooling among agent users suggests a maturity gradient: some users who engage deeply with agentic systems also adopt governance tooling. The governance challenge is to make that maturity gradient organizational rather than individual. Observability should not depend only on whether a local team happens to know the relevant tool stack.

The BTOS function-level results also complicate a purely technical governance response. The most common function in the national estimates is sales and marketing, followed by strategy and business development, information technology, and research and development. These are not all controlled by the same enterprise technology function, and they do not face the same risks. A marketing use may raise brand, copyright, or customer-segmentation issues. Strategy use may raise confidentiality and decision-quality issues. IT use may raise code, security, and operational dependency issues. R&D use may raise intellectual-property and scientific-validity issues. This functional diversity is consistent with the argument for task-level disclosure and triage. A single AI policy cannot substitute for governance that understands task context.

For manuscript inference, the most important caution is that none of these

public datasets prove that a specific employee used an unapproved tool in a specific organization. The paper does not make that claim. Instead, Study 2 provides mechanism-oriented plausibility evidence for shadow pressure. When AI use is frequent, trust is mixed, exposure is concentrated, functions are diverse, and agentic tooling is present but unevenly instrumented, the conditions exist for meaningful AI-mediated innovation to occur outside formal portfolios. That is the exact situation for which an uncertainty observatory is designed. Study 2 therefore informs RQ2 by identifying public signals consistent with the adoption-governance gap and informs RQ3 by showing why observability, verification, and task-level triage must be built into governance.

6. Study 3: Incident weak signals and operational uncertainty

Study 3 examines public AI provider-status incidents as weak signals of operational uncertainty. The incident dataset is intentionally modest and bounded: 25 provider-status incidents across five providers during a short January 2026 window. It should not be interpreted as a comprehensive AI harm database or as an estimate of provider reliability rates. Its value lies elsewhere. Provider-status incidents reveal the kinds of operational disturbances that can affect both formal and shadow GenAI workflows: degraded APIs, partial outages, authentication problems, integration failures, latency, and model-quality or behavior signals. These are the infrastructures on which everyday AI-mediated work depends.

The coding results show concentration around operational fragility. The top category is api errors availability, accounting for 16 of 25 incidents. The top provider in this short feed is Anthropic with 15 incidents. The most common uncertainty signal is partial degradation, present in 20 incidents. Partial degradation is especially important because it is harder to govern than a binary outage. A system that is obviously down triggers fallback routines. A system that is slower, intermittent, or producing elevated errors can generate ambiguous workarounds: employees retry prompts, switch models, copy data into another tool, bypass internal gateways, or accept lower verification standards under deadline pressure. Table 11 summarizes the coding scheme and observed counts.

Table 11: Operational incident taxonomy and uncertainty-observatory response

Failure category	Incidents	Major outages	Median hours	Observatory response
api errors availability	16	7	1.08	Runtime sensing, user-facing degradation flags, fallback routes
other uncoded	5	1	2.55	Open anomaly review and post-incident taxonomy update
authentication access	1	1	0.46	Access revocation playbook and credential event correlation
data pipeline integration	1	0	0.57	Dependency mapping and data-lineage alerts
latency performance	1	0	4.56	Performance baselines and throttling/queue policies
model quality behavior	1	0	2.63	Output sampling, regression tests, human review escalation

Uncertainty signal counts across incident titles/status metadata: partial degradation=20; technical opacity=4; unknown or investigating=4; external dependency=1. The feed is used as operational weak-signal evidence, not as a complete AI-harm database.

In Figure 8, we map provider exposure, category concentration, duration, and uncertainty-signal profiles.

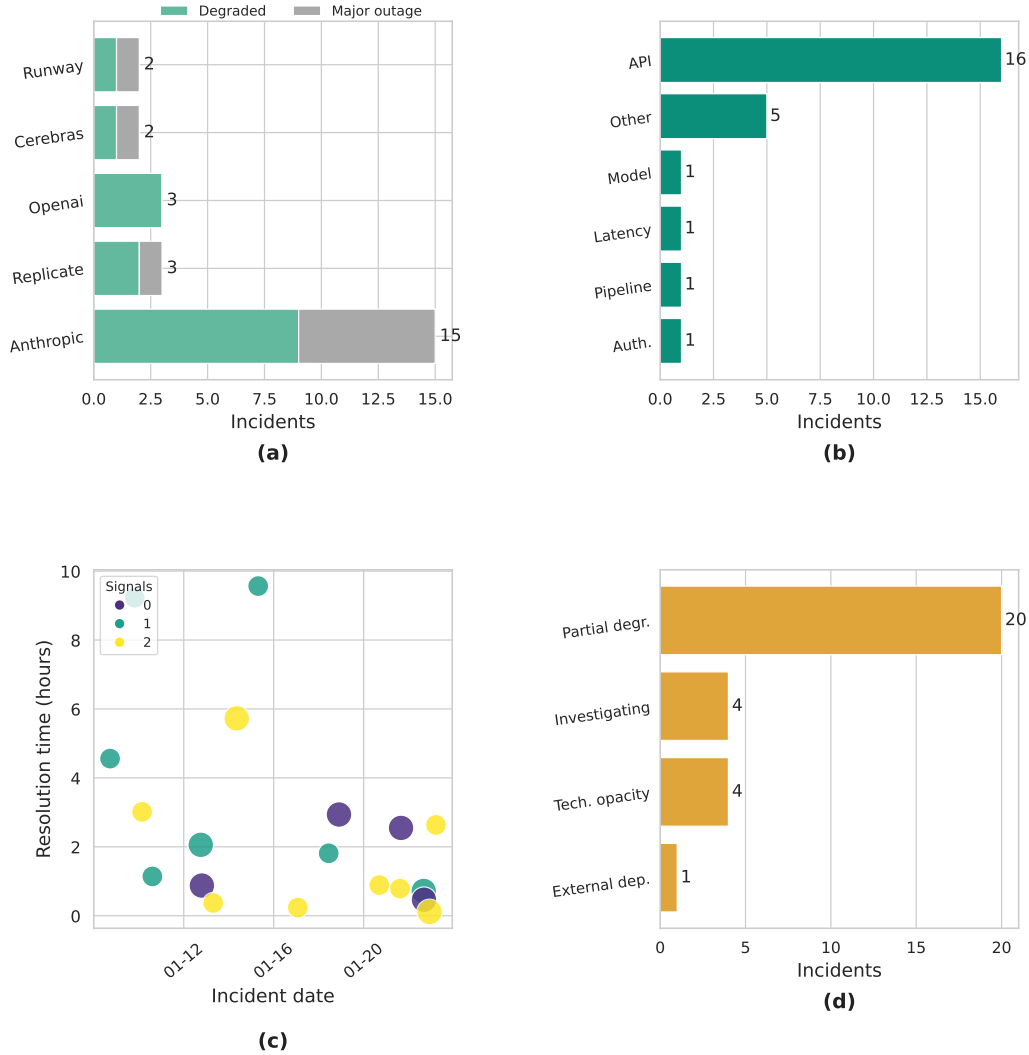


Figure 8: Operational weak-signal map from public AI provider-status incidents: (a) provider exposure and severity; (b) coded failure categories; (c) duration and uncertainty intensity; and (d) uncertainty signal profile. Categories and uncertainty signals are rule-coded from public status metadata; descriptions were not available in the raw feed.

The incident analysis strengthens the paper’s distinction between risk registers and uncertainty observatories. A risk register can list third-party service outage as a risk. It can assign an owner, record a mitigation, and specify fallback procedures. But the weak-signal problem is more dynamic. Which teams are affected by a partial degradation? Which unofficial tools do

they switch to? Do they paste sensitive data into alternate services? Does an agent retry failed calls automatically? Are generated artifacts reviewed differently when the preferred model is unavailable? Are incident lessons incorporated into policy? These questions concern organizational learning under uncertainty, not merely risk recording.

The incident categories also show why shadow GenAI cannot be governed only at the point of model approval. Authentication/access incidents point to authority ambiguity: who or what is permitted to act? Data-pipeline/integration incidents point to boundary coupling: which downstream systems depend on AI outputs? Latency/performance incidents point to temporal pressure: when the system slows, do employees wait, bypass, or improvise? Model-quality/behavior incidents point to epistemic uncertainty: when output behavior changes, how do users know? The same incident can be low risk for a brainstorming task and high risk for a production deployment or compliance response.

Study 3 therefore contributes a weak-signal vocabulary. The point is not that public provider incidents reveal internal shadow workflows directly. These events do not show that shadow workarounds occurred. They identify operational conditions under which workarounds become plausible and governance-relevant. An uncertainty observatory should capture these signals, connect them to internal task contexts, and convert repeated patterns into updated risk categories. Without such a capability, organizations may learn about GenAI failure modes only after they have already become incidents, policy violations, or reputational events.

The short incident window is a limitation, but it also illustrates why weak-signal governance matters. A mature risk program may already know that vendors can experience outages. The more subtle issue is that GenAI workflows often degrade rather than fail cleanly. A developer may receive slower completions, partial code suggestions, or intermittent API errors. A support team may see summarization quality vary. A browser agent may retry a task without making the retry visible to the user. These events can produce local adaptations before any formal incident is declared internally. Employees may switch providers, disable safeguards, copy content into another model, or postpone verification in order to meet deadlines; the public incident feed does not observe those behaviors, but it identifies the degradation conditions under which such workarounds become plausible and governance-relevant.

From a technical-governance perspective, the incident analysis points to three observability requirements. First, organizations need dependency

observability: which teams, tasks, and workflows depend on which AI services and model endpoints? Second, they need behavioral observability: when a model or agent produces degraded, delayed, or anomalous output, how is this detected and escalated? Third, they need workaround observability: when a preferred service fails, what alternate tools do employees use? Conventional IT incident management tends to focus on service restoration. Shadow GenAI governance must also ask what happened during degradation and whether compensating practices created new exposure.

The categories in Table 11 therefore operate as governance prompts. API errors/availability signals raise questions about fallback and retry behavior. Authentication/access incidents raise questions about permissions and identity. Data-pipeline/integration incidents raise questions about downstream dependencies. Latency/performance incidents raise questions about time pressure and bypass. Model-quality/behavior incidents raise questions about evaluation, output drift, and human review. Other uncoded signals remind organizations that the taxonomy itself must remain revisable. This is the central difference between a static register and an observatory: the latter expects its categories to change.

Incident evidence also helps prevent an overly individualistic account of shadow GenAI. Employees do not improvise in a vacuum. They improvise within infrastructures that are themselves unstable, opaque, and externally controlled. If a worker moves from an approved tool to an unapproved tool during a provider degradation, that behavior may be a response to a real operational constraint. This does not remove accountability, but it changes the governance response. The organization should ask why the workaround was attractive, whether approved fallback options existed, whether the task was time-critical, and whether the workaround revealed a missing capability. Blame alone is a poor learning mechanism under uncertainty.

For future empirical work, richer incident data would allow stronger analysis. Internal incident reports could connect provider events to employee behavior, data exposure, and downstream outcomes. Near-miss reporting could reveal cases where AI output was almost used incorrectly but caught by human review. Version-control and workflow logs could show whether generated code or agentic changes cluster around service disruptions. Such data are difficult to obtain, but the public incident study shows the analytical structure: classify the signal, connect it to task context, assess reversibility, and update governance. Study 3 informs RQ2 by showing operational weak signals that public adoption data cannot capture, and it informs RQ3 by spec-

ifying the observability and containment functions an uncertainty observatory must perform.

7. From risk registers to uncertainty observatories

The empirical studies and propositions point toward a governance architecture that complements rather than replaces risk registers. Risk registers are appropriate for known hazards. They are less appropriate as the primary instrument for shadow GenAI innovation because the relevant categories are often emerging. The proposed uncertainty observatory has six functions: sensing, safe disclosure, triage, boundary mapping, containment and reversibility, and institutional learning. Each function addresses a specific limitation of risk-register governance under high uncertainty, and each must be implemented with legitimacy safeguards so that observability does not become surveillance by another name.

The observatory should be understood as a governance capability, not as a single office, dashboard, or surveillance tool. It can be hosted by a chief AI officer, AI governance function, responsible AI board, risk function, or digital transformation office, but it only works if it connects legal/compliance, information security, data governance, HR/training, procurement, business units, worker representatives, and technical platform owners. It also differs from an AI center of excellence. A center of excellence often spreads expertise and reusable practices; an observatory learns from weak signals, near misses, task disclosures, and workarounds. It differs from a risk register because it operates before categories stabilize. It differs from an AI inventory because it maps task chains, not just systems. It differs from incident response because it studies near misses and degradation conditions, not only declared failures.

The first function is *sensing*. Sensing involves collecting weak signals from multiple sources: tool logs, model gateways, code repositories, help-desk tickets, incident reports, employee disclosures, procurement requests, browser-extension inventories, data-loss-prevention alerts, provider-status events, and near-miss reports. Sensing should be proportional and privacy-aware. The goal is not to monitor every prompt as a surveillance practice. The goal is to know where GenAI is materially shaping tasks, where data sensitivity is high, where agents can act across tools, and where employees experience policy friction.

The second function is *safe disclosure*. Employees often hide or under-report unofficial technology use when reporting channels are punitive, slow,

or irrelevant to task demands. An observatory therefore needs safe and low-friction disclosure mechanisms: lightweight task-use forms, office-hour consultations, rapid exception review, sandbox requests, and near-miss reporting that distinguishes learning from misconduct. Disclosure should ask what task was supported, what data were used, what output was relied upon, how verification occurred, and whether an agent or external tool took action. These questions produce governance-relevant knowledge without assuming that every local use is a violation.

The third function is *triage*. Not all shadow GenAI use deserves the same response. A seven-dimensional triage can classify cases by authorization, visibility, task criticality, data sensitivity, output reliance, reversibility, and agentic autonomy. Low-sensitivity brainstorming may require minimal documentation. High-sensitivity legal, medical, financial, security, or customer-impacting use requires stronger controls. Agentic autonomy raises the triage level because action, not only text, can cross boundaries. Triage converts diffuse uncertainty into prioritized governance attention.

The fourth function is *boundary mapping*. GenAI workflows cross boundaries among employees, models, data stores, vendors, code repositories, and downstream systems. Boundary mapping identifies the actors, tools, data, permissions, and outputs involved in a workflow. For agents, it should include tool permissions, external calls, memory, logging, rollback options, and human approval points. Boundary mapping is the bridge between shadow IT and responsible AI: it turns an informal work practice into an analyzable socio-technical arrangement.

The fifth function is *containment and reversibility*. Because many GenAI failures are discovered after output has already circulated, governance must ask whether consequences can be contained or reversed. Generated code can be reverted if versioned; generated decisions may be harder to reverse if communicated to customers; leaked data cannot be recalled. Reversibility assessment should therefore be part of the approval and exception process. It should also inform incident response: when partial degradation or model behavior changes occur, organizations need fallback routines that prevent rushed migration to unapproved tools.

The sixth function is *institutional learning*. Observatories differ from dashboards because they close the loop. Repeated near misses should update policy. New tool categories should update inventories. Employee workarounds should reveal unmet needs. Incident patterns should update risk registers. Successful low-risk experiments should become approved patterns. In this way,

an uncertainty observatory gradually converts uncertainty into risk knowledge, but it does so through practice rather than through premature categorization. Table 12 translates these functions into input signals, responsible actors, artifacts, and governance actions.

Table 12: Uncertainty observatory operating model

Function	Input signals	Core actors	Artifact	Governance action	Learning output	Legitimacy safeguard
Sensing	Logs, tickets, code labels, procurement requests, DLP alerts, provider incidents, near misses	AI governance, security, data governance, platform owners	Weak-signal register	Identify patterns before stable risk categories exist	New candidate categories	Aggregate where possible; avoid full prompt capture by default
Safe disclosure	Task-use questions, exception requests, sandbox needs, reported workarounds	Business units, HR/training, legal, worker representatives	Task-use register and exception form	Surface useful experimentation without treating every disclosure as misconduct	Unmet tool needs and policy friction	Non-punitive channel except intentional harm or reckless disregard
Triage	Data sensitivity, task criticality, output reliance, reversibility, autonomy	Risk, legal, security, domain owner	Verification-burden matrix	Allocate review intensity proportionally to consequence	High-consequence task classes	Clear thresholds and right to contest classification
Boundary mapping	Model endpoints, agents, data stores, vendors, permissions, downstream systems	Platform owners, data stewards, process owners	Agentic boundary map	Assign accountability at tool, data, action, and decision crossings	Dependency and escalation map	Purpose limitation for records and access controls
Contain	Partial degradation, model-quality changes, failed verification, external dependency	Incident response, system owner, continuity, compliance	Reversibility checklist and fallback playbook	Decide whether output can be recalled, reverted, isolated, or revalidated	Fallback and rollback rules	Preserve evidence without turning near misses into routine discipline
Institutional learning	Repeated near misses, successful experiments, recurring friction, new tool categories	Responsible AI board, risk owners, digital leadership, business units	Policy update log and pattern catalogue	Convert uncertainty into risk entries, approved patterns, or retired controls	Updated register and use-pattern rules	Worker participation in pattern review and policy revision

The observatory complements existing risk registers, AI inventories, model-risk processes, procurement review, and incident response. Its distinctive role is learning from weak signals and task practices before categories stabilize while limiting observability to legitimate governance purposes.

7.1. Design principles

Three design principles follow from the empirical analysis. The first is proportional visibility. Organizations should not treat every GenAI interaction as requiring the same level of monitoring. Proportionality means that visibility requirements increase with data sensitivity, task criticality, output reliance, irreversibility, and agentic autonomy. A low-stakes brainstorming use may require only general guidance. A workflow that generates customer-facing advice, modifies code, or acts across systems requires logging, review, and accountability. Proportional visibility is important because excessive monitoring can drive work underground, while insufficient monitoring makes learning impossible.

The second principle is psychological safety for disclosure. Shadow practices persist when employees believe disclosure will be punished, ignored, or slowed by bureaucracy. An observatory should separate learning-oriented disclosure from misconduct investigation except in clearly harmful cases. This does not mean that anything is allowed. It means that the organization creates a channel where employees can ask whether a tool can be used for a task with a particular data class and receive a timely answer. Near misses should be treated as learning assets. A near miss that is reported is an opportunity to update governance; a near miss that is hidden becomes latent risk.

The third principle is purpose limitation and contestability. Signals collected for learning, near-miss prevention, or policy updating should not be silently reused for productivity ranking, individual discipline, or generalized monitoring. Employees need to know which task classes are visible, who can access reports, how long records are retained, and how they can explain a workaround when approved tools were unavailable or inadequate. This principle is what distinguishes an uncertainty observatory from simply expanding managerial observation.

The fourth principle is boundary accountability. GenAI workflows often cross from human to model, employee to vendor, approved to unapproved tool, internal to external data, suggestion to action, and draft to decision. Accountability should be assigned at boundary crossings rather than only at tool ownership. For example, an AI agent that opens pull requests crosses from recommendation to action. A summarization tool used on customer data crosses from internal context to model-mediated processing. A generated analysis used in a board presentation crosses from draft support to decision

influence. Boundary accountability asks who approved the crossing, what evidence supports it, and how consequences can be reviewed or reversed.

7.2. Implementation roadmap

An organization can implement an uncertainty observatory in stages. The first stage is a minimum viable observatory. It inventories approved AI tools, names accountable owners, opens a non-punitive disclosure channel, and collects a first set of near misses and exceptions. The second stage is instrumentation. It adds lightweight logging, model-gateway controls, code-repository labels, data-loss-prevention alerts, and incident tags where appropriate. The third stage is boundary mapping. It identifies high-exposure task chains, agent permissions, external vendors, data classes, and downstream systems. The fourth stage is the learning loop. It reviews disclosures and incidents periodically, updates approved patterns, retires ineffective controls, and escalates repeated weak signals into formal risk categories. The fifth stage is institutionalization. It connects the observatory to risk registers, AI inventories, model-risk processes, procurement, training, and incident response.

This staged approach is intentionally pragmatic. Many organizations cannot begin with a complete AI inventory because informal use is already underway. They can begin by asking where AI is creating work value, where employees experience policy friction, and where output verification is difficult. They can then prioritize high-consequence workflows. The observatory is therefore not a large centralized bureaucracy. It is a learning infrastructure that connects local knowledge to enterprise governance.

7.3. Relationship to risk registers

The relationship between observatories and risk registers is complementary. Risk registers are artifacts of known risk. Observatories are capabilities for discovering and stabilizing emerging risk. When an observatory identifies a recurring pattern, the pattern can be translated into a risk-register entry with an owner, control, and review cycle. When a risk-register control generates workarounds, the observatory can detect the workaround and revise the control. This two-way relationship is important. Without observatories, risk registers become stale. Without risk registers, observatories may generate signals without accountability. Together, they create a governance loop.

The loop has a temporal logic. Before a category exists, the observatory senses weak signals. As patterns recur, governance names the category and

tests controls. Once controls stabilize, the risk register records them. When the environment changes, the observatory detects drift. GenAI requires this temporal logic because model capabilities, vendor products, employee practices, and regulatory expectations are all changing. Governance must therefore be built for revision. In Figure 9, we summarize the observatory as a learning-oriented governance capability.

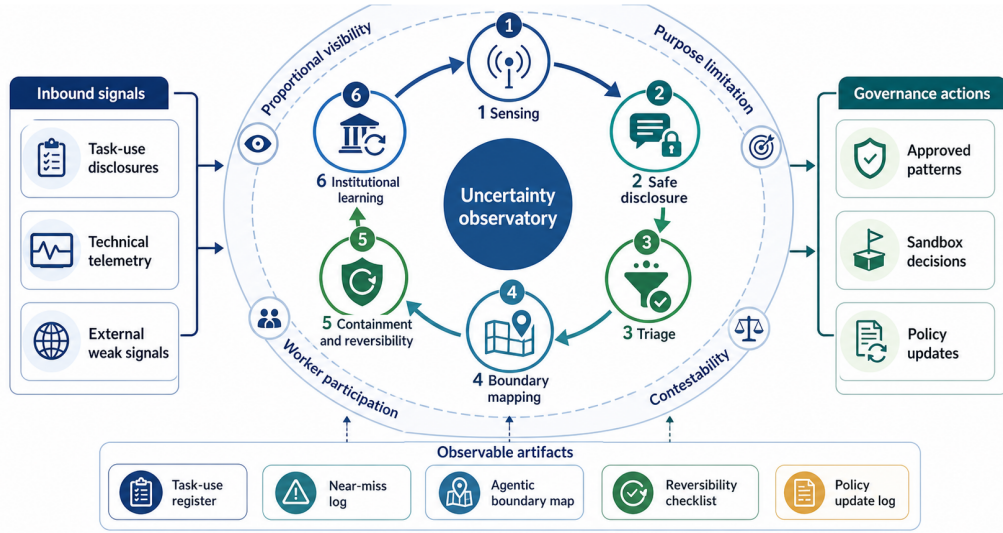


Figure 9: Uncertainty observatory as a learning-oriented governance capability. The figure summarizes the movement from weak-signal sensing and disclosure to triage, boundary mapping, containment, and institutional learning.

In Figure 10, we position observatory-oriented governance against three familiar alternatives. Repressive mode minimizes visible risk through prohibition and strict control; its benefit is clear accountability for known hazards, but its risk is driving useful experimentation underground, making it most appropriate for prohibited data classes or safety-critical tasks such as externally hosted processing of regulated records. Permissive mode encourages local experimentation; its benefit is speed and user learning, but its risk is value slippage and weak accountability, making it appropriate only for low-sensitivity brainstorming or sandboxed exploration. Integrative mode absorbs useful practices into formal governance; its benefit is scaling what works, but its risk is moving too slowly when categories are still forming,

making it appropriate for recurring low-to-moderate-risk practices such as approved prompt-pattern libraries. Observatory-oriented mode treats weak signals, disclosures, incidents, and near misses as learning inputs; its benefit is proportional control under uncertainty, and its main risk is legitimacy loss if implemented as surveillance rather than safe disclosure. The paper, therefore, does not recommend permissiveness. It recommends observatory-oriented governance because it combines proportional control with institutional learning.

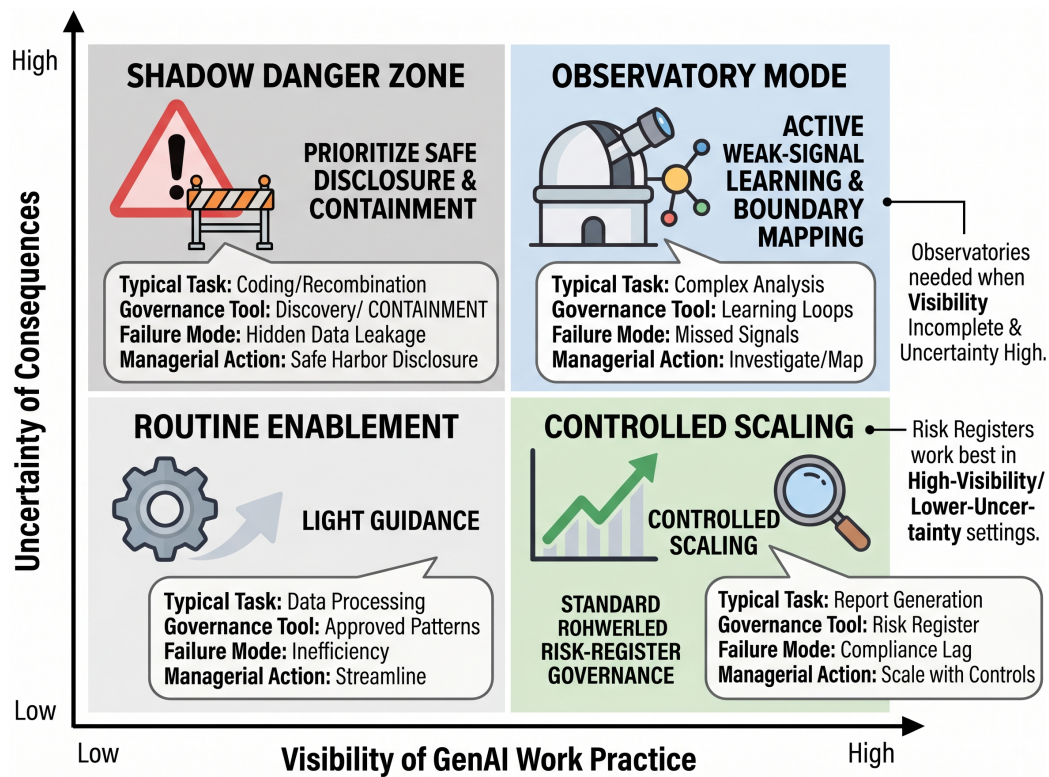


Figure 10: Governance mode matrix for shadow GenAI innovation. The matrix distinguishes repressive, permissive, integrative, and observatory-oriented modes by their treatment of risk tolerance and learning orientation.

8. Discussion

This paper began from a practical puzzle: GenAI can generate organizational value precisely because it is easy to use locally, but that same ease makes it difficult to govern. The public evidence map is consistent with the premise

that organizations face conditions consistent with an adoption-governance gap. BTOS shows firm-level diffusion across sectors and functions. Stack Overflow shows widespread developer use, mixed trust, agentic adoption, and uneven observability/security tooling. Anthropic exposure data show that AI pressure is concentrated in particular occupations and tasks. The structured literature synthesis shows that governance and GenAI research are active but not yet consolidated around task-chain shadow GenAI. Incident data show operational weak signals around API availability, partial degradation, access, and integration. Together, these findings make the case for treating shadow GenAI innovation as a high-risk and high-uncertainty governance problem while avoiding prevalence claims that public data cannot support.

8.1. Theoretical contributions

The first theoretical contribution is a task-level definition of shadow GenAI innovation. This definition extends, rather than replaces, the closest prior streams. Chin et al. (2025) show that shadow AI usage already matters for knowledge leakage and sustainability, and Waters-Lynch et al. (2025) show that covert GenAI use can be employee-led innovation rather than simple misuse. The present paper adds a task-chain account: GenAI requires a shift from system authorization to practice visibility. The same approved tool can support low-shadow or high-shadow use depending on task criticality, data sensitivity, output reliance, reversibility, and autonomy. This reframing matters because it prevents two errors. The first error is over-control: treating all informal GenAI use as deviance and thereby suppressing useful local learning. The second error is under-control: treating approved-tool access as sufficient governance even when the task remains high stakes and poorly verified.

Shadow AI usage and shadow user innovation can therefore be understood as close predecessors and, in some cases, special instances within the broader task-chain uncertainty account developed here.

The second theoretical contribution is the distinction between risk-register and uncertainty-observatory governance. Risk registers are not rejected. They remain necessary for known risks and accountability. The problem is that shadow GenAI often begins before risks are known enough to categorize. An uncertainty observatory provides the missing learning function. It captures weak signals, interprets them in task context, maps boundaries, supports containment, and updates risk categories. This distinction extends dynamic

capabilities theory by specifying a governance capability for converting distributed work-practice uncertainty into institutional knowledge.

The third theoretical contribution is the proposition-driven operating model in Table 12. The model connects diffusion pressure, category lag, verification burden, agentic boundary shifts, constraint shadowing, weak-signal learning, and legitimate observability to concrete governance artifacts. These mechanisms explain why shadow GenAI is not merely a compliance failure. It emerges from the interaction of useful tools, task pressure, incomplete governance, and uncertainty. This framing aligns with the social organization of technology: the relevant question is how societies and organizations shape technology use while being reshaped by it.

8.2. Methodological contribution

The paper also offers a methodological contribution for studying hidden or partially visible technologies. Direct measurement is difficult when the phenomenon is partly concealed. Rather than overclaim, the study triangulates public proxies and states their inference limits. This approach is useful for emerging technologies where organizational logs are proprietary, practices are in flux, and public datasets observe adjacent signals rather than the construct itself. The result is not causal identification of shadow behavior; it is a transparent mechanism-based evidence map. Future researchers can extend this design with organization-level surveys, ethnography, secure telemetry, diary studies, incident-report analysis, and matched governance-policy data.

8.3. From uncertainty management to uncertainty regulation

The argument also reframes uncertainty management as uncertainty regulation. Managing uncertainty can imply a local managerial response to ambiguity. Regulating uncertainty, in the sense used here, means building institutional arrangements that decide which weak signals count, which disclosures are protected, which boundary crossings require review, and which recurring patterns become formal risk categories. This matters for digital ecosystems because GenAI use is distributed across vendors, platforms, employees, and professional communities. No single organization controls all relevant conditions, but organizations can regulate how uncertainty enters their own routines.

8.4. Managerial implications

For managers, the central implication is that banning or approving tools is not enough. GenAI governance must move closer to tasks. Organizations should ask where employees use AI, what data are involved, how outputs are verified, whether agents can act across tools, and whether mistakes are reversible. They should distinguish low-risk experimentation from high-stakes reliance. They should create safe disclosure channels so employees can reveal workarounds without immediate punishment. They should track AI-related friction, not only AI access, because friction often predicts where shadow practices emerge.

The observatory model also suggests practical governance artifacts. A task-level AI disclosure form can be short and useful if it asks for task, data type, tool, output reliance, verification, and agentic action. A GenAI boundary map can identify which systems and vendors a workflow touches. A near-miss log can capture almost-wrong outputs, failed prompts, model behavior changes, and policy workarounds. A reversibility checklist can ask whether generated outputs can be recalled, reverted, or audited. An exception review board can approve temporary experiments while requiring logging and review. These artifacts are not bureaucratic additions for their own sake. They are ways to turn hidden practice into governable learning.

8.5. Digital leadership, skills, and value slippage

The framework has direct implications for digital leadership and skills. Leaders need to cultivate a capability that is neither naive experimentation nor blanket restriction. They must ask where approved tools fail to meet work needs, where employees carry verification burden, and where the organization lacks enough digital skill to distinguish productive experimentation from dangerous reliance. Digital skills therefore include not only prompt writing or model selection, but also evidence checking, data-boundary reasoning, agent-permission reasoning, and escalation judgment. These skills are organizational capabilities because they determine whether GenAI creates value, leaves value missing through suppressed experimentation, or produces value slippage through hidden dependencies and unaccountable outputs.

8.6. Policy implications

At the societal level, shadow GenAI raises questions about accountability and legitimacy. If AI-mediated work happens outside formal systems, affected stakeholders may not know when AI contributed to a decision, output, or

service interaction. Regulators and standards bodies increasingly emphasize risk management, transparency, and accountability, but organizational reality includes informal use, consumer tools, and distributed experimentation. Policy should therefore encourage not only formal AI inventories but also mechanisms for near-miss reporting, employee protection when raising concerns, incident transparency, and documentation of high-impact AI-mediated work practices. The public interest is not served by forcing all experimentation underground.

These policy implications do not imply that regulators or employers should monitor everyday prompts. Rather, public standards can encourage organizations to maintain disclosure channels, auditability expectations, near-miss learning, and documentation for high-impact AI-mediated work. This proportional approach protects worker legitimacy and psychological safety while preserving governance for consequential AI-mediated tasks.

8.7. Societal implications: observability, legitimacy, and worker cost

The observatory metaphor raises an unavoidable societal concern: being observed can itself impose costs. A governance capability designed to make weak signals visible can become a workplace surveillance system if it is implemented without limits, participation, and due process. Scholarship on worker surveillance and algorithmic management warns that monitoring systems can normalize asymmetric visibility, discipline workers through data traces, and shift discretion from people to managerial analytics (Ajunwa et al., 2017; Kellogg et al., 2020). In the GenAI context, these concerns are especially sharp because the traces are close to cognition and craft: prompts, drafts, code suggestions, verification notes, near misses, and exploratory reasoning. If employees believe that every AI-assisted experiment may be inspected as evidence of incompetence or disloyalty, they may self-censor, avoid useful experimentation, hide mistakes, or withhold the very weak signals governance needs.

This is where an uncertainty observatory must be distinguished from generic “better management.” Better management is a broad aspiration: clearer policies, more training, better tools, and more attentive supervision. An uncertainty observatory is a bounded governance capability with explicit artifacts, thresholds, and protections. It specifies what must be disclosed, which task classes require review, which signals are collected in aggregate, when escalation is triggered, who can access the records, how long records are retained, and how near-miss reports are separated from discipline except in cases of intentional harm or reckless disregard. Its artifacts are concrete:

task-use registers, exception forms, near-miss logs, agentic boundary maps, verification-burden matrices, reversibility checklists, policy-update logs, and approved or prohibited use-pattern catalogues. Without those boundaries, the observatory collapses into ordinary managerial attention or, worse, into surveillance under a softer name.

The human cost of observability is not distributed evenly. Junior workers, contractors, remote employees, outsourced teams, and workers in highly monitored roles may experience AI-use reporting as a test of loyalty rather than as a learning channel. Professionals whose identities are tied to expertise may also hesitate to disclose AI assistance if disclosure signals replaceability, incompetence, or reduced professional judgment. The Work Trend Index evidence that many AI users bring their own tools to work and worry about how disclosure will be interpreted (Microsoft WorkLab, 2024) is therefore not only an adoption statistic; it is a legitimacy signal. Governance that lacks psychological safety will receive distorted data, because workers will report what feels safe rather than what is true. Psychological safety is not a cultural luxury in this setting; it is an epistemic condition for accurate observability (Edmondson, 1999).

A legitimate uncertainty observatory should therefore follow proportional visibility. Low-sensitivity brainstorming should not generate the same trace as AI-mediated legal advice, customer-impacting decisions, security changes, or agentic code modifications. Purpose limitation is equally important: information collected for learning, near-miss prevention, or policy updating should not be silently reused for productivity ranking, discipline, or individual performance scoring. Data minimization should favor task categories, data classes, and boundary crossings over full prompt capture unless the task is high consequence and employees know the rule in advance. Contestability should allow workers to explain why a workaround occurred, whether approved tools were unavailable, and whether the local practice revealed an unmet organizational need.

These safeguards do not weaken governance. They make it more credible. If workers trust that disclosure is bounded, proportionate, and oriented toward learning, the organization is more likely to see weak signals before they become incidents. If workers experience observability as coercive, shadow practices will become more hidden, more individualized, and less governable. The societal implication is that responsible GenAI governance depends not only on technical controls but also on the legitimacy of organizational observation. An uncertainty observatory is different from better management precisely

because it makes this legitimacy problem part of the design: visibility must be earned, scoped, and reciprocated through better tools, safer reporting, and institutional learning.

8.8. Limitations

In our current study, the literature study is structured and extensive but bounded: it uses transparent search, screening, topic-modeling, co-occurrence, and closest-predecessor procedures, but it is not a dual-coded exhaustive full-text systematic review. The Stack Overflow survey is self-reported and developer-centered, so it cannot represent all occupations or industries. BTOS provides representative firm-level estimates but not detailed internal governance practices. The Anthropic Economic Index reflects a platform-specific public trace and, in the local files used here, does not directly distinguish augmentation from automation. The incident feed is short-window provider-status data and lacks detailed narratives. The seven dimensions are a theoretical diagnostic rather than a validated psychometric scale. The paper does not estimate the prevalence of shadow GenAI, identify authorization status in organizations, or observe employee intent. These limits reflect the empirical difficulty of studying a partly hidden phenomenon. The paper therefore frames claims as proxy-supported and mechanism-oriented.

8.9. Boundary conditions

The argument has boundary conditions. First, shadow GenAI pressure varies by regulatory environment. Highly regulated sectors may have stronger formal controls, but they may also produce stronger incentives for hidden workarounds when approved tools lag behind task demands. Less regulated sectors may have weaker controls but more open experimentation. Second, the framework applies most strongly to knowledge work in which language, code, analysis, documentation, and decision support are central. GenAI can affect physical operations as well, but the mechanisms of visibility, reversibility, and agentic autonomy will differ when AI is embedded in hardware, robotics, or industrial control. Third, organizational culture matters. A culture that treats mistakes as learning opportunities may receive more accurate weak signals than a culture that punishes disclosure. Fourth, vendor architecture matters. Enterprise gateways, audit logs, model-routing controls, and data-boundary protections can reduce shadow pressure, but only if they meet work needs well enough to be used.

These boundary conditions do not weaken the observatory argument; they specify where it should be adapted. In a highly regulated organization, the observatory may emphasize compliance evidence, formal exception records, and audit trails. In a creative or research-intensive organization, it may emphasize experimentation sandboxes, prompt-pattern libraries, and rapid review. In software development, it may emphasize code provenance, repository labels, test coverage, and agent permissions. In customer operations, it may emphasize disclosure, escalation, quality review, and data minimization. The core capability remains the same: observe emerging use, interpret it in task context, and update governance before hidden routines become institutionalized.

The framework is most applicable where GenAI is accessible, recombinable, and capable of shaping knowledge work outputs. It is less directly applicable in environments where tools are technically locked down and work cannot be mediated through external models, although even those environments may face pressure through personal devices, contractors, or vendor systems. Safety-critical, defense, health-care, education, public-sector, and small-business contexts require adaptation because the balance among experimentation, evidence, accountability, and worker autonomy differs across institutional settings.

8.10. Future research

Our future research intends to test the propositions with organizational data. We believe surveys could measure the seven dimensions of shadow GenAI: authorization, visibility, task criticality, data sensitivity, output reliance, reversibility, and agentic autonomy. Similarly, ethnographic studies could examine how employees decide when to disclose or conceal their use of GenAI, while field experiments could compare punitive policies with safe-disclosure observatories. In addition, incident studies could connect provider outages to internal workaround behavior. Longitudinal research could examine how organizations convert weak signals into updated risk registers over time. Finally, comparative research could examine how sector, professional norms, and regulation shape the balance between innovation and governance.

A second research path concerns measurement. The public-data design in this paper deliberately avoids collapsing heterogeneous signals into a single index, but organization-level studies could build and validate such measures. An adoption-governance gap index could compare task-level AI use with the presence of approved tools, documentation, verification routines,

and observability. A shadow pressure index could combine policy friction, external tool availability, task urgency, data sensitivity, and lack of approved alternatives. An uncertainty signal intensity index could combine incident ambiguity, partial degradation, model drift, and near-miss frequency. These measures would allow future research to test whether observatory practices reduce hidden use, improve reporting, or increase responsible scaling.

A third research path concerns power and worker experience. Observability can support responsible governance, but it can also become surveillance. The observatory concept should therefore be studied critically. Researchers should ask when disclosure systems feel safe, when employees perceive them as punitive, how managers use reported data, and whether observability disproportionately burdens certain roles. The related scholarship is well-positioned to examine these normative tensions because GenAI governance is not only an efficiency problem. It concerns trust, autonomy, accountability, and the legitimacy of organizational control.

The practical implication is that uncertainty observatories should be designed with worker participation. Employees who use GenAI in everyday tasks often know where the real verification burden lies. They know which approved tools are insufficient, which outputs are misleading, which workflows are time-critical, and which policies create incentives for concealment. Governance that excludes those employees will be less accurate and less legitimate. Governance that includes them can convert shadow knowledge into institutional learning.

9. Conclusion

GenAI has made innovation easier to start and harder to see responsibly. This paper argues that the resulting governance problem is not only that GenAI creates new risks, but that it creates uncertainty about where innovation is happening, which tasks are affected, how outputs are verified, when agents cross boundaries, and who is accountable when consequences emerge. Shadow GenAI innovation names this condition: work-use of GenAI systems, agents, or model-enabled workflows that can create value while remaining only partly authorized, visible, instrumented, or accountable.

The empirical analyses support the urgency of the issue while keeping the paper’s claims bounded. Public firm-level evidence indicates meaningful AI diffusion across sectors and functions. Developer survey evidence shows high AI use, mixed trust, visible agent adoption, and uneven observability/security

tooling. Occupation and task exposure data show concentrated pressure on particular work domains. The structured literature synthesis shows that existing scholarship has not yet consolidated shadow GenAI as a theoretical object. Incident data show operational weak signals that can turn ordinary AI-mediated work into ambiguous failure. None of these streams directly measures hidden behavior, authorization status, or employee intent. Together, they support a mechanism-oriented interpretation of the conditions that can make partially governed use plausible and governance-relevant.

The paper's main proposal is uncertainty observatory governance. Risk registers remain necessary for known hazards, but they need a companion capability for weak-signal sensing, safe disclosure, triage, boundary mapping, containment, reversibility, and learning. The practical goal is not to eliminate local GenAI experimentation. It is to make experimentation visible enough to learn from, safe enough to scale responsibly, and accountable enough to maintain trust. For organizations navigating high risk and high uncertainty, the move from risk registers to uncertainty observatories may determine whether GenAI becomes a source of responsible innovation or a growing field of hidden exposure.

The argument is deliberately balanced. Shadow GenAI is not treated as inherently reckless, and formal governance is not treated as inherently wise. Both local experimentation and centralized control can create value or harm depending on how they are organized. The contribution is to show that GenAI makes this balance harder because valuable experimentation can happen before categories, controls, and accountabilities stabilize. Organizations, therefore, need governance that is empirical, adaptive, and close to work practice. They need to see enough to learn without driving practice underground.

In technology and society, the broader lesson is that emerging technologies are governed not only by standards, policies, and formal systems but also by the everyday arrangements by which people get work done. GenAI is now part of those arrangements. The question is whether organizations can turn distributed experimentation into responsible learning before hidden exposure accumulates. Uncertainty observatories offer one answer: build institutions that can notice weak signals, protect disclosure, map boundaries, and revise governance as the technology and its uses continue to evolve.

Declaration of generative AI use

During manuscript preparation, the authors used ChatGPT for language refinement, structural editing, and drafting support. The authors reviewed and edited the output and take full responsibility for the content of the published article.

Data availability statement

The original public data sources used in this study include the U.S. Census Bureau Business Trends and Outlook Survey, Stack Overflow Developer Survey, Anthropic Economic Index, OpenAlex, and public AI provider-status incident data. The processed research materials, analysis scripts, search-query documentation, aggregate results, generated tables and figures, codebook, checksums, excluded-data notes, and supplementary replication artifacts are available in an external repository. Repository details have been provided separately through the journal submission system to preserve double-anonymized review.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Search protocol and structured literature synthesis

The literature search used the OpenAlex Works API plus prior OpenAlex/Crossref seed metadata. The enriched OpenAlex pass used the following exact query strings: “shadow AI generative AI governance”; “shadow IT generative artificial intelligence organizations”; “generative AI governance enterprise risk management”; “large language models responsible AI governance organizations”; “digital innovation uncertainty governance”; “responsible AI risk management uncertainty”; “agentic AI governance risk”; “AI agents governance organizational risk”; “foundation models enterprise governance risk”; “generative AI workplace adoption risk governance”; “AI innovation uncertainty management”; and “algorithmic governance uncertainty digital innovation”. The OpenAlex Works API filtered records from 2015-01-01 to 2026-12-31, requested 50 records per query, sorted by citation count, and selected fields for DOI, title, year/date, venue, authors, concepts, abstract, citation count, and type. An older seed script also searched six shorter query strings from 2020 onward; those records were retained as prior metadata and deduplicated with the enriched pass.

The pipeline retained query provenance, DOI, title, venue, year, abstract where available, concepts, authors, citation count, and OpenAlex identifiers. No separate execution timestamp was stored in the output files; the relevant evidence audit was produced on 2026-04-26. Records were deduplicated using DOI first and normalized title as fallback. When duplicates appeared, the retained record favored abstract availability and higher citation count. Screening used a transparent rule-based score over title, abstract, concepts, and venue; query strings were retained for provenance but were not counted as substantive evidence for inclusion. Inclusion required a connection among

AI/GenAI and at least one organizational, work, governance, risk, uncertainty, innovation, or agentic-system signal. Broad biomedical, clinical, or general uncertainty matches without organizational AI relevance were removed. The final structured literature corpus retained 53 records from 401 deduplicated records, with abstracts available for 52 records.

Because screening was implemented as a rule-based single-pipeline procedure rather than independent human coding, no inter-rater reliability statistic is reported. This choice is consistent with the paper’s use of the corpus as a transparent evidence map and theory-building synthesis rather than as a dual-coded systematic review.

Topic modeling used title, abstract, and concept text. The TF-IDF vectorizer used one- and two-word n-grams, English stopwords plus domain-specific stopwords, `min_df=2`, `max_df=0.9`, and a letter-token regex. The non-negative matrix factorization model used five topics for the 53-record corpus, `random_state=42`, `init=nndsvda`, and `max_iter=600`. The environment specification pins Python 3.10 but does not pin every Python package version; the reproducibility package should therefore record installed package versions before public archiving.

The search is best understood as a structured, search-based literature synthesis rather than an exhaustive full-text systematic review. The analysis combines bibliographic records, available abstracts, concepts, topic modeling, keyword co-occurrence, and targeted closest-predecessor reading. The targeted predecessor reading included Chin et al. (2025) on shadow AI usage and Waters-Lynch et al. (2025) on shadow user innovation because these 2025 articles directly shape the contribution boundary even if their appearance at the edge of the fast-moving literature should not retroactively alter the retained corpus counts. The article therefore reports the review process transparently while constraining its claims to conceptual concentration, vocabulary gaps, and theory-building positioning.

Appendix B. Variable dictionary

Table B.13: Key variables used in the empirical analysis.

Variable family	Source	Operationalization
Current firm AI use	BTOS	Percentage estimate for current AI use in any business function.
AI function use	BTOS	Percentage estimates for business functions such as sales and marketing, strategy, IT, R&D, customer service, and legal/compliance.
AI-tool use	Stack Overflow 2025	Binary and ordinal indicators from current use frequency: any, monthly/infrequent, weekly, daily, plan, and no plan.
AI-output trust	Stack Overflow 2025	Positive trust equals somewhat or highly trust. Nonnegative trust equals neutral, somewhat trust, or highly trust. Ordered trust score used in robustness.
Verification burden	Stack Overflow 2025	AI friction, frustration counts, perceived complex-task capability, and human fallback items.
Agent use	Stack Overflow 2025	Binary and ordinal indicators for daily, weekly, monthly/infrequent, planned, copilot-only, and no-plan responses.
Agent governance stack	Stack Overflow 2025	Observability/security tool use, orchestration tool use, external agent/copilot use, and broad governance-stack robustness indicator.
Occupation exposure	Anthropic Economic Index	Observed exposure by occupation and task penetration, reported as proportions and exposure bands.
Incident category	AI Status Dashboard	Rule-based coding of provider-status incident titles/status into availability, access, integration, latency, model-quality, and uncoded operational categories.
Uncertainty signal	AI Status Dashboard	Rule-based coding for partial degradation, technical opacity, unknown/investigating status, and external dependency.

Appendix C. Model and robustness details

The Stack Overflow models use item-specific valid denominators. Logistic models estimate any AI use, positive AI-output trust, AI-agent use, and observability/security stack use among agent users using maximum 200 iterations and HC3 robust standard errors. Ordered-logit models estimate AI-use frequency, trust score, and agent-use frequency. Covariates include role group, country group, professional developer status, employment/contractor status, work experience, people-manager status, large-organization status, remote/hybrid work, and advanced degree where available. Robustness checks use alternate dependent-variable thresholds: weekly/daily AI use, daily AI use, nonnegative trust, weekly/daily agent use, agent planned-or-used, and broad agent governance stack.

The main models are associational. They are used to identify whether adoption intensity, trust, agent use, and governance tooling move together in ways consistent with the adoption-governance-gap mechanism. They should not be interpreted as causal estimates of individual behavior.

M1 estimates current AI-tool use among respondents with the required covariates. M2 estimates positive AI-output trust among AI users and includes use intensity, frustration, friction, and perceived complex-task capability. M3 estimates AI-agent use and adds AI-use intensity and trust. M4 estimates observability/security stack use among agent users and includes agent-use frequency, orchestration tools, external agents, security/privacy concern, and strict-rule challenge indicators. These model-specific samples and predictors are why Equation (1) is presented as a template rather than a single common specification.

Appendix D. Additional contextual industry signals

Table D.14 reports practitioner, vendor, and public-data statistics that helped motivate the practical salience of the adoption–governance gap. These sources are heterogeneous in sampling design and are not part of the three-study empirical design. They are retained here as contextual signals only.

Table D.14: Additional contextual signals on AI use, shadow use, data exposure, and governance capacity

Source	Metric	N/base	Use	Shadow	Data	Gov.	Risk
NCA/Cyb.	AI use	n=6,500+	65%; +21 pp	–	–	58%	–
NCA/Cyb.	Sensitive input	AI users	–	–	43%; 50%; 42%; 44%	–	–
ME	Worker use	n=350 W	–	93%; 82%; 60%	–	–	–
ME	Exposure	350 W; 350 IT	–	–	32%; 37%	91%; 54%	85%
TELUS	Public assistants	n=1,000	68%	–	57%	24%; 44%	–
TELUS	Input types	n=1,000	–	–	31%; 29%; 21%	–	–
WalkMe	Shadow use	n=1,000	78%	49%	–	–	80%
WalkMe	Training	n=1,000	–	–	–	7.5%; 23%; 51%	–
1Password	Access gap	n=5,200	73%	25%; 52%	–	33%	–
1Password	Device	n=5,200	–	–	22%; 43%	–	–
IBM	AI breach	n=600 orgs	–	20%	–	97%; 63%	13%; USD 670k
IBM	Audit coverage	AI-policy orgs	–	–	–	34%	–
Microsoft WTI	Worker use	n=31,000	75%	78%	–	60%	–
McKinsey	Org adoption	survey	88%; 71%	–	–	–	6%
Cisco	Readiness	n=8,000	–	–	–	24%; 19%	13%; 15%

Note: Cells report percentages unless another unit is shown. Multiple entries in a cell are compact source-reported submetrics in the same category; dashes indicate that the source does not report that quantity. ME = ManageEngine; Gov. denotes policy, training, readiness, or observability capacity. These statistics are heterogeneous contextual signals rather than comparable estimates or direct prevalence measures of hidden conduct. Source citations: National Cybersecurity Alliance and CybSafe (2025); ManageEngine (2025); TELUS Digital (2025); WalkMe (2025); 1Password (2025); IBM (2025); Microsoft WorkLab (2024); McKinsey & Company (2025); Cisco (2025).